

1 Earlier endpoint proof with propagated signed control

1.1 The two endpoint calculations give a four-volume threshold of four

1.1.1 Result and motivation

The arithmetic endpoint note bounds the monic source norms. The subsequent window-product calculation retains the repeated occurrence of every interior volume contraction. Composing them on the window one degree shorter than the original window improves the original four-volume lower threshold from 2 to 4:

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k-1} V_{q_k} / (V_{2q_k-1} V_{2q_k}))}{q_k \log k} \geq 4.$$

This statement concerns the original arithmetic family belonging to a fixed packet consisting exactly of a full off-line quartet, with every multiplicity retained. It is a necessary volume-growth consequence for such a packet, not an assertion that an off-line zero exists. No upper bound contradicting this growth is established here.

This note is a written continuation of the two supplied calculations. It is not included in the frozen Toda–Gamma edition DOI [10.5281/zenodo.22730756](https://doi.org/10.5281/zenodo.22730756).

1.1.2 1. Original source, constants, and coordinates

Keep $g = 2\xi$, the fixed monic packet polynomial h , and

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad M_h(b) = \int e^{bt} w_h(t) dt.$$

The mass remains μ_h^k . The monic degree- j source norm is $\omega_{h,k,j}$ for the observation $S = k/2 + iu$. In this coordinate map the leading coefficient acquires the phase i^j , of squared modulus one. Neither the source measure nor the generator is replaced.

The arithmetic endpoint proof gives constants $c_h > 0$, $\vartheta_h = \int_{-1}^1 w_h > 0$, and $B_h = 42 + 2 \deg h$, such that

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\pi(|u|+k-3)/2} (1 + |u| + k - 3)^{-B_h}, \quad k \geq 3. \quad (1)$$

Here (1) is the proved lower envelope of the actual threefold-and-higher convolution, not a newly postulated comparison measure. Its proof is in the [supplied complete arithmetic note](#).

Choose one $0 < b < \pi/2$ and define bounding constants

$$X_h = \max\{1, M_h(b), M_h(-b)\}, \quad K_h = \max\{1, \vartheta_h^{-1}\}.$$

Taking maxima here only bounds constants; it performs no rescaling of the source, its quotient, or either Gram determinant.

1.1.3 2. The balanced-window norm estimate, with an explicit constant

For integers $n \geq k \geq 3$ and $n/2 \leq r \leq n$, one has

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq C_h^{\text{bal}} n, \quad (2)$$

where the following literal finite constant suffices:

$$C_h^{\text{bal}} = 256e^\pi \max\{1, 3/c_h\} X_h K_h \max\{1, b^{-3}\} 6^{B_h/3}. \quad (3)$$

To prove this, the trial polynomial $(S - k/2)^j$ and exponential moments give the unchanged upper estimate

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k).$$

The exact monic Legendre norm on $[-n, n]$ is

$$\mathfrak{l}_n(n) = \frac{2n^{2n+1}}{2n+1} \left(\frac{2^n}{\binom{2n}{n}} \right)^2 \geq \frac{2n^{2n+1}}{(2n+1)4^n}.$$

This follows from the Legendre leading coefficient and squared norm in [DLMF Table 18.3.1](#). Combining (1) with this least-norm calculation gives the exact finite bound

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq \left[\frac{(2(n+r))! b^{-2(n+r)} (M_h(b)^k + M_h(-b)^k) e^{\pi(n+k-3)/2} (1+n+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2r)}. \quad (4)$$

For completeness, use $(2(n+r))! \leq (4n)^{2(n+r)}$, $M_h(b)^k + M_h(-b)^k \leq 2X_h^k$, $\vartheta_h^{-(k-3)} \leq K_h^k$, and $1+n+k-3 \leq 2n$. After taking the root, the factorial and Legendre factors are bounded by

$$4n 8^{n/r} \left(\frac{2n+1}{c_h n} \right)^{1/(2r)}.$$

The remaining factors are

$$b^{-(n+r)/r} (X_h K_h)^{k/(2r)} e^{\pi(n+k-3)/(4r)} (2n)^{B_h/(2r)}.$$

Now $n/r \leq 2$, $(n+r)/r \leq 3$, $k/(2r) \leq 1$, and $(n+k-3)/(4r) \leq 1$. Also $2n+1 \leq 3n$, and $\log(2n)/n$ decreases for $n \geq 3$, so $(2n)^{B_h/(2r)} \leq 6^{B_h/3}$. These inequalities prove (2)–(3). The diagonal $r = n$ recovers the scale of the supplied theorem. The needed new case is $r = n - 1$, which lies in the same balanced range.

1.1.4 3. The exact product on the same canonical metrics

Let $V_N = \det G_N$, $\delta_N = V_N/V_{N-1} \in (0, 1]$, and retain the calculated real phase $\phi_N = \sigma - \partial_\theta \log V_N$. The original radius identity is

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N) (\delta_{N+1}^{-1} - 1), \quad N \geq q.$$

Multiply at $N = n, \dots, n+r-1$, without discarding intermediate factors:

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} (1 - \delta_n) (1 - \delta_{n+r}) \prod_{j=1}^{r-1} (1 - \delta_{n+j})^2. \quad (5)$$

The original phase remains on the left. Each factor on the right after the ratio belongs to $[0, 1]$. The exterior theorem bounds the aggregate spectral defect $L_{h,k}$ by every ϵ_N . Thus (5) proves

$$L_{h,k}^{2r} \leq \frac{\omega_{h,k,n+r} V_n}{\omega_{h,k,n} V_{n+r}}. \quad (6)$$

This deduction needs neither a new metric nor an assumption that the phase vanishes. A unit contraction makes a factor in (5) zero and directly forces $L_{h,k} = 0$; the inequality (6) still holds. The first admitted degree $q - 1$ is not used in any radius identity here.

1.1.5 4. Central window and the factor four

For the exact quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$ and common order m , keep

$$q_k = [1 + k(m - 1)](k + 1)^2, \quad L_{h,k} = 2\delta[1 + k(m - 1)](k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor.$$

In particular $L_{h,k}/q_k \geq \delta k/2$. Put $q = q_k$ and choose $n = q, r = q - 1$. All hypotheses of the proved balanced-window estimate are satisfied: $q \geq k \geq 3, q - 1 \geq q/2$, and every radius index is at least q . Equations (2) and (6) yield

$$\log \frac{V_q}{V_{2q-1}} \geq 2(q - 1) \log \frac{L_{h,k}}{C_h^{\text{bal}} q} \geq 2(q - 1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (7)$$

Write $C_k = \log(V_q/V_{2q-1})$. The original four-volume budget is exactly

$$\begin{aligned} \mathcal{B}_{h,k} &= \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} \\ &= 2C_k + \log \frac{V_{q-1}}{V_q} + \log \frac{V_{2q-1}}{V_{2q}} \geq 2C_k. \end{aligned} \quad (8)$$

Both last terms are nonnegative by the original rank-one kernel update. Substitution of (7) proves

$$\mathcal{B}_{h,k} \geq 4(q_k - 1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (9)$$

1.1.6 The signed metric control at these same four endpoints

The AW calculation supplies the full-spectrum estimate and its condition-number consequence. The SP calculation supplies the additional actual-overlap refinement. Their exact comparison ACM.1–13 proves that they act on the same source. We carry both calculations into the budget (8), rather than replacing its four signed terms by four separate errors. Here the integration coordinate remains the original u in $S = k/2 + iu$; the coordinate called y in ACM is exactly this u . The new variable $x \in [0, 1]$ is only the metric interpolation parameter.

Put $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and $\mathcal{H} = \mathcal{P}_{2q}$ in its original ordered monomials, $v(u) = (1, S, \dots, S^{2q})$, and retain the full arithmetic density $m_1(u) = w_h^{*k}(u)$. The original Gamma comparison has the literal density and mass

$$\begin{aligned} r_a(u) &= \frac{|\Gamma(a + iu/2)|^2}{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a), \\ m_0(u) &= \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}(u), \quad c_{1/4} = \sqrt{2\pi}, \quad \int_{\mathbb{R}} m_0(u) du = (\sqrt{2\pi})^k. \end{aligned} \quad (\text{FV.M1})$$

The complete Gamma convolution proof retained in this reader gives $m_0 = r_{1/4}^{*k}$. Thus (FV.M1) uses its original mass; the arithmetic mass is still μ_h^k . Define

$$m_x = (1-x)m_0 + xm_1, \quad M_x = \int_{\mathbb{R}} v(u)^* v(u) m_x(u) du, \quad C_x = M_x^{-1}(M_1 - M_0). \quad (\text{FV.M2})$$

Both endpoint densities are positive almost everywhere and have all moments, so every M_x is positive: a nonzero polynomial has only finitely many zeros on the integration line. No matrix or density is divided by its mass.

Let $I_N : \mathbb{C}[S]_{\leq N} \hookrightarrow \mathcal{H}$ be coefficient inclusion and $B_N : \mathbb{C}[S]_{\leq N-q} \rightarrow \mathbb{C}[S]_{\leq N}$ be multiplication by the original monic χ . Negative-degree spaces are zero. Put $H_N(x) = I_N^* M_x I_N$, and use the original remainder J_N to set $G_N(x) = (J_N H_N(x)^{-1} J_N^*)^{-1}$ and $V_N(x) = \det G_N(x)$. The maps to the common source and their orthogonal projections are exactly

$$\begin{aligned} \tilde{B}_N &= I_N B_N, & P_N &= I_N H_N(x)^{-1} I_N^* M_x, \\ Q_N &= \tilde{B}_N (\tilde{B}_N^* M_x \tilde{B}_N)^{-1} \tilde{B}_N^* M_x, & Q_{q-1} &= 0. \end{aligned} \quad (\text{FV.M3})$$

The formula for Q_{q-1} uses the zero relation domain, with empty determinant one. For each other N , injectivity of multiplication by monic χ proves invertibility of its relation Gram. Direct multiplication proves $P_N^2 = P_N$, $Q_N^2 = Q_N$ and self-adjointness in M_x , with ranges $\mathcal{P}_N$ and $\chi \mathcal{P}_{N-q}$. These formulas are also the exact typed identities $\tilde{B}_N = B_N^{\text{SP}} = I_N B_N^{\text{AW}}$.

The coefficient matrix $[1, S, \dots, S^{q-1}, \chi, S\chi, \dots, S^{N-q}\chi]$ is triangular with diagonal one. Its Gram has determinant $\det H_N(x)$. Eliminating the relation block leaves its Schur complement $G_N(x)$: indeed for a fixed remainder z , minimizing over the relation coefficients solves the relation normal equations and gives precisely $z^* G_N(x) z$. Consequently

$$\det G_N(x) = \frac{\det H_N(x)}{\det(B_N^* H_N(x) B_N)}, \quad \frac{d}{dx} \log V_N(x) = \text{Tr}((P_N - Q_N) C_x). \quad (\text{FV.M4})$$

For the derivative, multilinearity of determinant gives $d \log \det H = \text{Tr}(H^{-1} dH)$. Insert $dH_N = I_N^* (M_1 - M_0) I_N$ in the numerator and $B_N^* dH_N B_N$ in the denominator. Cyclicity of the finite trace and $M_x C_x = M_1 - M_0$ give the two terms in (FV.M4).

Define the original signed function and its two projection sums by

$$\begin{aligned} F(x) &= \log \frac{V_{q-1}(x) V_q(x)}{V_{2q-1}(x) V_{2q}(x)}, \\ U_x &= Q_{2q-1} + Q_{2q} - Q_q, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q), \quad A_x = U_x - W_x. \end{aligned} \quad (\text{FV.M5})$$

Keeping the signs of all four terms in (FV.M4) gives

$$\begin{aligned} F'(x) &= \text{Tr}(A_x C_x) = \int_{\mathbb{R}} (m_1(u) - m_0(u)) v(u) A_x M_x^{-1} v(u)^* du, \\ \mathcal{B}_{h,k} &= F(1) = \mathcal{B}_{h,k}^{\Gamma} + \Delta_{h,k}, \quad \mathcal{B}_{h,k}^{\Gamma} = F(0), \quad \Delta_{h,k} = \int_0^1 F'(x) dx. \end{aligned} \quad (\text{FV.M6})$$

The scalar density uses the full inverse M_x^{-1} , not an ordinary Euclidean projection kernel. The integral identity follows by substituting $M_1 - M_0 = \int v^* v (m_1 - m_0)$ in the finite trace; its entries are integrable moments. The density is real because $A_x M_x^{-1}$ is Hermitian. It is signed, and its m_x integral equals $\text{Tr} A_x = 0$, as proved next.

The relation subspaces for $q, 2q-1, 2q$ are nested, of dimensions $1, q, q+1$. Hence U_x has eigenvalue two on $\chi \mathcal{P}_{q-1} \ominus \chi \mathcal{P}_0$ (dimension $q-1$), eigenvalue one on $\chi \mathcal{P}_0$ and $\chi \mathcal{P}_q \ominus \chi \mathcal{P}_{q-1}$ (one dimension

each), and eigenvalue zero on its q -dimensional complement. The degree flag decomposes W_x identically: its two summands have overlap $\mathcal{P}_{2q-1} \ominus \mathcal{P}_q$ of dimension $q-1$, and two one-dimensional end pieces. Thus both U_x, W_x have rank $q+1$, trace $2q$ and trace of the square $4q-2$. These statements include $q=1$, when the overlap dimension $q-1$ is zero.

Let $c_1(x) \leq \dots \leq c_{2q+1}(x)$ be the eigenvalues of C_x in M_x , let $d_x = c_{2q+1}(x) - c_1(x)$, and put $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. Retain both exact pointwise expressions

$$\begin{aligned} S_{\text{AW}}(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ S_{\text{SP}}(x) &= \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}, \\ \mathfrak{I}_{h,k} &= \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\} dx. \end{aligned} \tag{FV.M7}$$

Here is the full finite proof of their applicability at this earlier endpoint. For a rank- r orthogonal projection E , its diagonal in an orthonormal eigenbasis of C_x lies in $[0, 1]$ and sums to r . Moving this diagonal mass to the lowest or highest eigenvalues gives $\sum_{j=1}^r c_j \leq \text{Tr}(EC_x) \leq \sum_{j=2q+2-r}^{2q+1} c_j$. Each of U_x, W_x is its range projection, of rank $q+1$, plus its eigenvalue-two projection, of rank $q-1$. Applying the two inequalities to these two summands and subtracting the resulting extrema gives $|F'(x)| \leq S_{\text{AW}}(x)$, with the multiplicities displayed in (FV.M7).

Dimension gives $s_x \geq 1$. Let E_x be the M_x -orthogonal projection onto the actual intersection of the two ranges. Each of U_x, W_x dominates its own range projection, which dominates E_x ; therefore $U_x - E_x$ and $W_x - E_x$ are positive and each has trace $2q - s_x$. The triangle inequality for their trace norms gives $\|A_x\|_1 \leq 4q - 2s_x$. The full spectra just proved also give

$$\text{Tr } A_x = 0, \quad \text{Tr } A_x^2 = 8q - 4 - 2\text{Tr}(U_x W_x), \quad \|A_x\|_1^2 \leq (2q+1) \text{Tr } A_x^2. \tag{FV.M8}$$

The last inequality is Cauchy–Schwarz on all $2q+1$ real eigenvalues. Subtract $(c_1 + c_{2q+1})I/2$ from C_x in the trace; the first identity leaves F' unchanged. In an eigenbasis of A_x , each diagonal entry of the centered C_x has absolute value at most $d_x/2$. Thus $|F'(x)| \leq d_x \|A_x\|_1/2 \leq S_{\text{SP}}(x)$. Both estimates hold at every x on the same original metric. Their pointwise minimum is integrable: all Grams and inverse Grams are smooth, all ordered eigenvalues are continuous, and s_x is Borel since its level sets are determined by vanishing and nonvanishing matrix minors. Consequently (FV.M6) gives the finite two-sided enclosure

$$\boxed{\mathcal{B}_{h,k}^\Gamma - \mathfrak{I}_{h,k} \leq \mathcal{B}_{h,k} \leq \mathcal{B}_{h,k}^\Gamma + \mathfrak{I}_{h,k}.} \tag{FV.M9}$$

For the positive original relative endomorphism $D = M_0^{-1}M_1$ on $(\mathcal{H}, M_0)$, let $b_1 \leq \dots \leq b_{2q+1}$ be all its eigenvalues. Direct factorization $M_x = M_0((1-x)I + xD)$ gives $C_x = ((1-x)I + xD)^{-1}(D - I)$ and hence

$$\begin{aligned} c_j(x) &= \frac{b_j - 1}{1 - x + xb_j}, \quad \int_0^1 c_j(x) dx = \log b_j, \\ \mathfrak{I}_{h,k} &\leq \min \left\{ \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \quad \int_0^1 S_{\text{SP}}(x) dx \right\} \leq (2q-1) \log \frac{b_{2q+1}}{b_1}. \end{aligned} \tag{FV.M10}$$

The scalar fraction is increasing in b with derivative $(1 - x + xb)^{-2}$, proving the stated eigenvalue order throughout the path. Integrating its logarithmic derivative proves the integral. For the last bound use $s_x \geq 1$ in (FV.M7) and integrate d_x . The common consequence is the earlier AW bound; the pointwise minimum retains both its spectral refinement and the additional SP overlap.

The recursive metric transport proof RMT1–22 also applies here on the actual moment source and supplies a further propagated refinement. Its coordinate map is $y \mapsto u = y$, leaving $S = k/2 +$

iu unchanged. Equation (FV.M1) equals its reference measure (RMT2)–(RMT3), including $c_{1/4}^k = (\sqrt{2\pi})^k$; the arithmetic measure is identically $w_h^{*k} du$ in both. Therefore all entries of its M_x equal (FV.M2). Its included relation matrix is $I_N B_N = \tilde{B}_N$ in (FV.M3), and its quotient is this same J_N . Substitution proves equality of both Gram inverses, every P_N, Q_N , and the four volumes individually. In particular its $\mathcal{B}(x)$ equals the original $F(x)$ in (FV.M5), with no reversal of an endpoint or alteration of a sign. Its quantities E_q, O_q, d_x are precisely S_{AW}, S_{SP}, d_x here.

The measure hypothesis of that result holds literally, rather than being assigned to an arbitrary coefficient form. Its required positivity $F(x) > 0$ can also be checked in these coordinates: if $F(x) = 0$, the two crossed nested ratios $(q-1, 2q)$ and $(q, 2q-1)$ both have determinant ratio one. Their least-norm sections, included in the common $\mathcal{H}$ by I_i, I_j , then agree, because $G_i - G_j = (I_i R_i - I_j R_j)^* M_x (I_i R_i - I_j R_j)$ is positive and determinant one forces equality of the two forms. In the first crossed pair the literal remainder section contains 1, which would be orthogonal to $\chi \mathcal{P}_q$. The polynomial $\chi^{\#k}(S) = \sum_j \bar{a}_j (k-S)^j$ for $\chi(S) = \sum_j a_j S^j$ equals $\overline{\chi(S)}$ on the same line. Orthogonality to $\chi \chi^{\#k}$ would force $\int |\chi(S)|^2 m_x(u) du = 0$, contrary to the positive degree- $2q$ moment Gram. This retains the full original relation polynomial.

The further two entries in the current recursive control are calculated on these same operators. Retain

$$\begin{aligned} T_q(x) &= \frac{d_x}{2} \|U_x - W_x\|_1, \\ \Sigma_x &= \text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{2q+1}, \\ S_{HS}(x) &= \sqrt{\Sigma_x \text{Tr}((U_x - W_x)^2)}. \end{aligned} \tag{FV.M15a}$$

The midpoint-centering argument proving (FV.M8) already proves $|F'(x)| \leq T_q(x) \leq S_{SP}(x)$ before either trace-norm majorant is taken. Both inequalities are retained. For the second new entry, C_x is self-adjoint in M_x because $M_x C_x = M_1 - M_0$ is Hermitian. Its centered operator $C_x - \text{Tr}(C_x)I/(2q+1)$ has squared Hilbert–Schmidt norm exactly $\Sigma_x \geq 0$, by expanding its square and trace. Since $\text{Tr}(U_x - W_x) = 0$, replacing C_x by this centered operator leaves the trace pairing $F'(x)$ unchanged. Cauchy–Schwarz on the finite matrix entries in an M_x -orthonormal basis therefore gives $|F'(x)| \leq S_{HS}(x)$, with the other squared norm exactly the trace in (FV.M8). This basis is used only to prove the Hilbert-space inequality: both endpoint forms and every source-coordinate operator remain the original ones. The expressions in (FV.M15a) equal RMT10a–RMT10b after the coefficient identifications already proved above.

Put $a = 2q - 1$. The full-angle estimate of RMT11, whose two complete crossed angle lists and original zero angle are retained in that proof, therefore gives the simultaneous quantities

$$\begin{aligned} A_q(x) &= a \sqrt{1 - e^{-F(x)/a}} d_x, \\ \mathfrak{J}_{h,k} &= \int_0^1 \min\{S_{AW}(x), T_q(x), S_{SP}(x), S_{HS}(x), A_q(x)\} dx \leq \mathfrak{J}_{h,k}, \\ \mathfrak{J}_{h,k}^{\text{nl}} &= \int_0^1 \min\left\{d_x, \frac{S_{AW}(x)}{a \sqrt{1 - e^{-F(x)/a}}}, \frac{T_q(x)}{a \sqrt{1 - e^{-F(x)/a}}}, \right. \\ &\quad \left. \frac{S_{SP}(x)}{a \sqrt{1 - e^{-F(x)/a}}}, \frac{S_{HS}(x)}{a \sqrt{1 - e^{-F(x)/a}}} \right\} dx \leq \log \frac{b_{2q+1}}{b_1}. \end{aligned} \tag{FV.M15}$$

The exact source identities above identify all five retained estimates individually with RMT9–RMT13, including the exact trace-norm and centered Hilbert–Schmidt entries. Each bounds the same $|F'(x)|$, so their pointwise minimum does also. In particular it is at most the earlier three-entry minimum and the two-entry $\mathfrak{J}_{h,k}$; none of their full proofs is discarded. Strict positivity and compactness of the fixed path keep the displayed denominators nonzero. RMT13 proves measurability of the actual range intersection, and (FV.M10) computes the last integral of d_x . Integrating gives $|\Delta_{h,k}| \leq \mathfrak{J}_{h,k}$ directly.

For completeness the scalar coordinate used to propagate this refinement is $\mathcal{H}_a(B) = 2 \operatorname{arccosh}(e^{B/(2a)})$. It has derivative $1/(a\sqrt{1 - e^{-B/a}})$ and inverse $2a \log \cosh(z/2)$, by direct differentiation and substitution on the positive real branches. Apply this derivative to the five bounds in (FV.M15) and integrate, exactly as in the same-source RMT17, to obtain

$$|\mathcal{H}_a(\mathcal{B}_{h,k}) - \mathcal{H}_a(\mathcal{B}_{h,k}^\Gamma)| \leq \mathfrak{I}_{h,k}^{\text{nl}}. \quad (\text{FV.M16})$$

The constructed signed interval (ISM31)–(ISM37), also propagated in (RMT19)–(RMT22), is on these identical original forms. Its frequency y is this u , its endomorphism X_x is exactly $C_x = M_x^{-1}(M_1 - M_0)$, and its $\mathcal{A}_x$ is exactly $U_x - W_x = A_x$ by (FV.M3)–(FV.M6). Its included relation matrix is $I_N B_N$ and its minimum section is $I_N R_N(x)$: both have the same remainder, are orthogonal to the same relation range, and uniqueness of the positive minimum proves equality. The bare symbol $C : E \rightarrow \mathcal{H}$ in ISM denotes the literal remainder lift; it is not the endomorphism C_x here. Thus its scalar correction $\mathcal{C}_{\text{ISM}} = \mathcal{B}(M_1) - \mathcal{B}(M_0)$ equals $\Delta_{h,k}$, with the original Gamma-to-arithmetic orientation.

Fix an integer $p_N \geq 0$ solely as a finite truncation order and set $L_{\text{ISM}} = L_{\text{RMT}} = p_N$. This symbol changes none of the original polynomial cutoffs N , spectral quantity $L_{h,k}$, or parameters k, u, x . Retain the original positive eigenvalues $b_1, \dots, b_n$ of $D = M_0^{-1}M_1$, where $n = 2q + 1$, and define the exact finite operators

$$\begin{aligned} B_x^\circ &= (1 - x)I + xD, & \alpha_x &= \frac{2(1 - x) + x(b_1 + b_n)}{2}, \\ H_x^\circ &= I - B_x^\circ/\alpha_x, & C_x^{[p_N]} &= \alpha_x^{-1} \sum_{r=0}^{p_N} (H_x^\circ)^r (D - I), \\ R_x^{[p_N]} &= (H_x^\circ)^{p_N+1} C_x, & E_x^{[p_N]} &= R_x^{[p_N]} - \frac{\operatorname{Tr} R_x^{[p_N]}}{n} I. \end{aligned} \quad (\text{FV.M16a})$$

Multiplying the finite geometric sum by $I - H_x^\circ$ gives $C_x - C_x^{[p_N]} = R_x^{[p_N]}$. Every factor is a real function of D and self-adjoint in the original M_x metric. Trace zero of A_x therefore gives the exact signed decomposition $F'(x) = \operatorname{Tr}(C_x^{[p_N]} A_x) + \operatorname{Tr}(E_x^{[p_N]} A_x)$. Define the actual center and residual, preserving their signs, by

$$\begin{aligned} j_{p_N} &= \int_0^1 \operatorname{Tr}(C_x^{[p_N]} A_x) dx, \\ e_{p_N} &= \int_0^1 \sqrt{\operatorname{Tr}((E_x^{[p_N]})^2) \operatorname{Tr}(A_x^2)} dx, \\ \vartheta &= \frac{b_n - b_1}{b_n + b_1}, & K_D &= \sum_{j=1}^n \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \end{aligned} \quad (\text{FV.M16b})$$

They are exactly J_{p_N}, ϵ_{p_N} of ISM36 and j_{p_N}, e_{p_N} of RMT20. Cauchy-Schwarz applied to the last trace in the exact decomposition and then integration proves

$$j_{p_N} - e_{p_N} \leq \Delta_{h,k} \leq j_{p_N} + e_{p_N}, \quad 0 \leq e_{p_N} \leq \vartheta^{p_N+1} \sqrt{K_D(8q - 6)} \rightarrow 0. \quad (\text{FV.M16c})$$

For the stated bound, the eigenvalues of H_x° have absolute value at most $\vartheta < 1$, as the midpoint of the extreme eigenvalues $1 - x + xb_1$ and $1 - x + xb_n$ is precisely α_x . The original eigenvalues of C_x are $(b_j - 1)/(1 - x + xb_j)$. Their absolute values are at most $|b_j - 1|/\min\{1, b_j\}$. Centering the residual subtracts the nonnegative square of its trace divided by n from its squared Hilbert-Schmidt norm. This proves $\operatorname{Tr}((E_x^{[p_N]})^2) \leq \vartheta^{2p_N+2} K_D$. Both positive U_x, W_x dominate their range projections, whose intersection has dimension at least one. Positivity of the trace product gives $\operatorname{Tr}(U_x W_x) \geq 1$, so (FV.M8) gives $\operatorname{Tr}(A_x^2) \leq 8q - 6$, proving (FV.M16c). All of these are continuous finite matrix integrands. Their exact integrals are the objects used here; a numerical integration

must retain its own integration error. Convergence is for this fixed pair of original source forms. It asserts no estimate uniform in k . The complete original-source and proper-support proof for this construction is (ISM38)–(ISM43) in Section 5. To compare its primitive with (FV.M12), instantiate only that primitive calculation with the ordered pair (M_1, M_x) . Its section difference is then exactly $\rho_{\text{ISM},1} - \rho_{\text{ISM},0} = I_N(R_N(x) - R_N(1))$, so its boundary has precisely the orientation in (FV.M12). At $x = 0$ this is the negative of the section difference for the original ordered pair (M_0, M_1) ; linearity multiplies its primitive and proper-source residual by -1 . This typed primitive comparison leaves the original Gamma-to-arithmetic orientation of $\Delta_{h,k}$ and the signed estimates above unchanged.

Write $H_0 = \mathcal{H}_a(\mathcal{B}_{h,k}^\Gamma)$ and retain the complete simultaneous endpoints

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{lo}} &= \max \left\{ 0, \mathcal{B}_{h,k}^\Gamma - \mathfrak{J}_{h,k}, 2a \log \cosh \left(\frac{\max\{0, H_0 - \mathfrak{J}_{h,k}^{\text{nl}}\}}{2} \right), \right. \\ &\quad \left. \mathcal{B}_{h,k}^\Gamma + j_{p_N} - e_{p_N} \right\}, \\ \mathcal{B}_{h,k}^{\text{hi}} &= \min \left\{ \mathcal{B}_{h,k}^\Gamma + \mathfrak{J}_{h,k}, 2a \log \cosh \left(\frac{H_0 + \mathfrak{J}_{h,k}^{\text{nl}}}{2} \right), \right. \\ &\quad \left. \mathcal{B}_{h,k}^\Gamma + j_{p_N} + e_{p_N} \right\}. \end{aligned} \tag{FV.M17}$$

The inverse is increasing and extends continuously by zero at zero. Applying it to (FV.M16) and intersecting the resulting interval with $|\Delta_{h,k}| \leq \mathfrak{J}_{h,k}$ and the oriented interval (FV.M16c) proves, for every fixed truncation order $p_N \geq 0$,

$$\mathcal{B}_{h,k}^{\text{lo}} \leq \mathcal{B}_{h,k} \leq \mathcal{B}_{h,k}^{\text{hi}} \leq \mathcal{B}_{h,k}^\Gamma + \mathfrak{J}_{h,k} \leq \mathcal{B}_{h,k}^\Gamma + \mathfrak{J}_{h,k}. \tag{FV.M18}$$

These formulas use the complete proof in Section 4, with the actual measure and coefficient correspondences just proved. The preceding generic signed matrix calculation (FV.M3)–(FV.M10) remains valid on its own stated domain; the full-angle refinement here is specialized to the actual original arithmetic/Gamma moment path.

Finally let the two unchanged arithmetic endpoint losses be $E_0 = \log(V_{q-1}/V_q)$ and $E_1 = \log(V_{2q-1}/V_{2q})$. Equation (8), with (FV.M6), gives the literal central identity and bounds

$$\begin{aligned} 2C_k &= \mathcal{B}_{h,k}^\Gamma + \Delta_{h,k} - E_0 - E_1, \\ \max \left\{ 0, \frac{\mathcal{B}_{h,k}^{\text{lo}} - E_0 - E_1}{2} \right\} &\leq C_k \leq \frac{\mathcal{B}_{h,k}^{\text{hi}} - E_0 - E_1}{2}, \\ 4(q-1) \log \frac{L_{h,k}}{C_h^{\text{bal}q}} + E_0 + E_1 &\leq \mathcal{B}_{h,k} \leq \mathcal{B}_{h,k}^{\text{hi}}. \end{aligned} \tag{FV.M11}$$

The first line is substitution in (8); the second uses (FV.M18) and $C_k \geq 0$; the third uses the first inequality in (7) without discarding either endpoint loss. All expressions retain the original full packet h , its exact χ , and the constants in (1)–(4). This gives an actual finite control for the old four-volume expression. It supplies no uniform estimate of $\mathfrak{J}_{h,k}$ as the packet order grows, so the original lower-growth conclusion (9) remains intact.

The comparison is induced by identity polynomial maps on every $\mathcal{P}_N$ and the identity on $\mathbb{C}[S]/(\chi)$. For the original source realization $\mathcal{V}P = P(\sum D_i)F_h^{\otimes k}$ and complete unit map $\eta[P] = [g/h]_h^{\otimes k} P(\sum s_i)$, these identities commute exactly with ηJ_N ; the values and every raw derivative of each polynomial are unchanged. The least lift at x is $R_N(x) = H_N(x)^{-1} J_N^* G_N(x)$, so $J_N(R_N(x) - R_N(1)) = 0$. Monic division gives a unique coefficient map $\kappa_N(x)$ with $R_N(x) - R_N(1) = B_N \kappa_N(x)$. Retain the original fixed-order tensor division $\chi(\sum s_i) = \sum_i h(s_i) Q_i(s)$. On a quotient vector z , the complete original theta primitive of this difference is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D)(\kappa_N(x)z)(\sum_j D_j)(F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \tag{FV.M12}$$

The tensor differential contributes the second sign $(-1)^{i-1}$ and $h(D_i)F_h = \Theta\phi_0$, so its boundary is exactly $\mathcal{V}(R_N(x) - R_N(1))z$. Thus the Gram comparison is attached to an explicit original source map and primitive; it does not identify the interpolated norm with the original arithmetic norm. For each unchanged outer label and independent coefficient face, retain the actual declared subcomplex $C_\lambda^\bullet \subseteq C_{\text{full}}^\bullet$ and its degree- $(k-1)$ source C_λ^{k-1} . Write Ξ for the primitive in (FV.M12). Whenever $d\Xi$ belongs to C_λ^k , its exact residual is $[d\Xi] \in C_\lambda^k/d(C_\lambda^{k-1})$. The full top-degree comparison has the exact kernel

$$\ker[H^k(C_\lambda^\bullet) \rightarrow H^k(C_{\text{full}}^\bullet)] = \frac{d(C_{\text{full}}^{k-1}) \cap C_\lambda^k}{d(C_\lambda^{k-1})}. \quad (\text{FV.M13})$$

Here both complexes have top degree k , so all degree- k elements are cycles. A class $[z]$ maps to zero precisely when $z = d\Xi$ for some $\Xi \in C_{\text{full}}^{k-1}$; it is already zero in the declared source precisely when $z = d\Xi_\lambda$ for a declared primitive. This proves (FV.M13), including its injective inclusion into $H^k(C_\lambda^\bullet)$. Equivalently the primitive map has domain $\{\Xi \in C_{\text{full}}^{k-1} : d\Xi \in C_\lambda^k\}$ and kernel $C_\lambda^{k-1} + \ker d$, so quotienting by this exact kernel gives the same residual space. This retains possible cancellations among tensor slots. For one leg with unchanged top source $\mathcal{B}$ and injective $\Theta : V \rightarrow \mathcal{B}$, it becomes the exact sequence

$$0 \rightarrow V/W \xrightarrow{[v] \mapsto [\Theta v]} \mathcal{B}/\Theta W \rightarrow \mathcal{B}/\Theta V \rightarrow 0. \quad (\text{FV.M14})$$

The first map is injective because $\Theta v \in \Theta W$ forces $v \in W$; its image is the kernel of the quotient map, which is surjective by definition. Thus a relation with primitive in the declared source maps to its supported zero. A general full-source primitive retains the exact residual (FV.M13) on every coefficient face, rather than declaring its proper-source class zero. Identity polynomial maps retain the outer labels; external τ still maps to τ and an empty coefficient face retains its outer label.

Since h is fixed and $q_k \rightarrow \infty$, division by $q_k \log k$ gives the claimed liminf of at least 4. An upper limit strictly below 4 would already conflict with (9); no such upper estimate has been obtained here. The two-volume conclusion at $n = r = q$ is also retained: $\liminf \log(V_q/V_{2q})/(q \log k) \geq 2$.

1.1.7 5. Source maps and verification scope

All representatives remain those of the supplied arithmetic construction: $J^{(k)}R_N = \eta_{h,k}$ and $q^{(k)}R_N = \sigma_h^{\otimes k} \eta_{h,k}$. Restricting the degree window does not alter their unit, jets, fibre labels, or cohomology classes. The supported-zero relation and the external absence τ retain their original separate maps. In particular, no repeated exterior iteration or erased multiplicity cost occurs in (5)–(9).

The analytic envelope and original norm theorem are from the supplied Arithmetic Endpoint Bounds note. Equation (5) is the subsequent supplied window-product calculation. Equations (2)–(4) and their composition (7)–(9) are the present continuation. The two independent review lanes record finite tests separately from written analytic proof; no new Lean certificate or interval enclosure of C_h^{bal} is asserted.

2 An arithmetic lower-envelope comparison for central quotient volumes: full jets, finite determinant bounds, and an exact quadratic benchmark

This calculation constructs upper bounds for the original central volume $\log(V_q/V_{2q-1})$. Its input is the proved lower envelope and Laplace estimate of the complete *Arithmetic Endpoint Bounds* source, equations (29)–(32). The comparison acts on the original polynomial and quotient maps, including all multiplicities and phases. The original matrix bound retains an explicit positive matrix: its arithmetic moments stop at degree $2q$, while its comparison moments continue to $4q-2$. A second bound uses only the supplied envelope constants and two Laplace values. The signed four-endpoint comparison is now propagated into (AU.31a)–(AU.31b); it retains the endpoint losses and uses actual

moments through degree $4q$. The explicit scalar estimate has quadratic, rather than logarithmic, growth per quotient direction. An exact auxiliary calculation determines that scale and supplies a monic family relating its full jets to the original quartet jets. No bound contradicting the quartet lower estimate is asserted.

2.1 Original source and every jet coordinate

Fix a complete quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$, and common complete zero order $m \geq 1$ of $g = 2\xi$. Keep its monic polynomial h , its entire quotient $v_h = g/h$, and

$$w_h(t) = |v_h(1/2 + it)|^2/(2\pi), \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad k \geq 3. \quad (\text{AU.1})$$

Write $c = k/2$ and keep the original observation $S = c + iu$. The source form is $\int \overline{P(c + iu)} Q(c + iu) m_k(u) du$, conjugate-linear in its first argument. Its constant norm is μ_h^k . The coefficient map and its inverse are

$$\Psi : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad P \longmapsto P(c + iu), \quad f \longmapsto f((S - c)/i). \quad (\text{AU.2})$$

These are inverse ring homomorphisms by substitution. Multiplication by S is carried to $c + iM_u$, and $\partial_u^d \Psi(P) = i^d \Psi(P^{(d)})$. On the ordered degree- N coefficient spaces the matrix is triangular with diagonal $1, i, \dots, i^N$, so its determinant has modulus one. The original quotient volumes therefore have exactly the same values in the transported remainder basis; no mass or norm is divided out.

Let $\ell = 1 + k(m - 1)$ and use the ordered distinct grid

$$\lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \zeta_{ab} = (\lambda_{ab} - c)/i = (2b - k)\gamma - i(2a - k)\delta, \quad 0 \leq a, b \leq k. \quad (\text{AU.3})$$

The original cyclic relation and its transported monic polynomial are

$$q = \ell(k + 1)^2, \quad \chi(S) = \prod_{a,b} (S - \lambda_{ab})^\ell, \quad \psi(u) = i^{-q} \chi(c + iu) = \prod_{a,b} (u - \zeta_{ab})^\ell. \quad (\text{AU.4})$$

The grid is invariant under complex conjugation in the u chart, so ψ is real monic. Equivalently the original coefficient identity is $\bar{\chi}(k - S) = (-1)^q \chi(S)$. In particular the original norm multiplier is $\bar{\chi}(c - iu) \chi(c + iu) = \psi(u)^2$, with the source multiplication still carrying the unit i^q .

For clarity the preceding cyclic degree also follows from the original tensor algebra. In a tensor of k length- m local factors, the sum of the k nilpotent generators has power $k(m - 1)$ nonzero: its coefficient on the product of their top powers is $(k(m - 1))!/((m - 1)!)^k$, nonzero in $\mathbb{C}$. Its next power is zero by total degree. Every grid sum in (AU.3) occurs, since the two signs in each quartet factor can be chosen independently. Direct products of local factors give the same maximal length ℓ at every grid sum. Distinct pairs have distinct real and imaginary parts. The minimal polynomial of the original sum is therefore exactly (AU.4), with no lost multiplicity.

Put $\mathcal{E} = \mathbb{C}[u]/(\psi)$, with ordered remainder coefficients $1, u, \dots, u^{q-1}$. For $N \geq q - 1$ let J_N be remainder and B_N multiplication by ψ in the ordered monomials. Monic division gives the exact sequence

$$0 \longrightarrow \mathbb{C}[u]_{\leq N-q} \xrightarrow{B_N} \mathbb{C}[u]_{\leq N} \xrightarrow{J_N} \mathcal{E} \longrightarrow 0. \quad (\text{AU.5})$$

The source on the left is zero if $N = q - 1$. Precisely, the coefficient isomorphisms are $\iota_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}[u]_{\leq N}$, $a \mapsto \sum_{j=0}^N a_j u^j$, and $\kappa : \mathbb{C}^q \rightarrow \mathcal{E}$, $x \mapsto [\sum_{j < q} x_j u^j]$. The polynomial quotient map is $\kappa J_N \iota_N^{-1}$ and the polynomial multiplication map is $\iota_N B_N \iota_N^{-1}$. When a matrix formula is evaluated on a polynomial below, these displayed coefficient isomorphisms are the specified identifications. The original S chart uses the same coefficient isomorphisms with u^j replaced by S^j . The complete raw derivative map is

$$\mathbf{E}_N f = (f^{(d)}(\zeta_{ab}))_{a,b,0 \leq d < \ell}, \quad (\mathbf{E}_N)_{(a,b,d),j} = \begin{cases} j! \zeta_{ab}^{j-d}/(j-d)!, & j \geq d, \\ 0, & j < d. \end{cases} \quad (\text{AU.6})$$

Taylor expansion at each distinct centre shows that its kernel is $(\psi) \cap \mathbb{C}[u]_{\leq N}$. Let $Z = E_{q-1}$. Then $E_N = ZJ_N$ and Z is invertible. Indeed a degree below q polynomial with all those zero raw jets is divisible by degree- q ψ and is zero. For the fixed row order,

$$\det Z = \left(\prod_{a,b} \prod_{d=0}^{\ell-1} d! \right) \prod_{(a,b) < (a',b')} (\zeta_{a'b'} - \zeta_{ab})^{\ell^2}. \quad (\text{AU.7})$$

One proof first regards all roots as distinct variables in the ordinary Vandermonde determinant, groups each cluster, and replaces its rows by successive divided differences. Each row operation divides by exactly the corresponding within-cluster difference. Sending the nodes in a cluster to its centre replaces its divided-difference row of order d by $f^{(d)}/d!$, by the finite Taylor expansion of each monomial. Multiplying those rows by $d!$ proves (AU.7). This is an identity of polynomials after removal of the indicated factors; no source norm or zero multiplicity is approximated. In the original S coordinates the same raw jet of order d has its literal factor i^d from (AU.2).

For any positive source matrix H_N , the least quotient form is

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \quad R_N = H_N^{-1} J_N^* G_N. \quad (\text{AU.8})$$

Surjectivity proves $K_N > 0$. Multiplication proves $J_N R_N = I$ and $B_N^* H_N R_N = 0$. Thus every lift $R_N x + B_N y$ has norm $x^* G_N x + y^* B_N^* H_N B_N y$, proving the minimum property. In raw jets the kernel is $K_N = E_N H_N^{-1} E_N^*$, and $\det K_N = |\det Z|^2 / V_N$. Hence

$$C_k := \log(V_q / V_{2q-1}) = \log \det K_{2q-1} - \log \det K_q. \quad (\text{AU.9})$$

All factorials in (AU.7) cancel only in this specified ratio.

2.2 The lower envelope as an explicit positive comparison form

The supplied arithmetic proof establishes $c_h > 0$, $\vartheta_h = \int_{-1}^1 w_h > 0$, $B = 42 + 2 \deg h = 42 + 8m$, and

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad \alpha = \pi/2. \quad (\text{AU.10})$$

Set $a_k = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}$ and $d_k = k - 2 > 0$. Define the comparison measure, retaining its literal mass, by

$$d\nu_k(u) = a_k e^{-\alpha|u|} (d_k + |u|)^{-B} du. \quad (\text{AU.11})$$

The identity map $L^2(m_k du) \rightarrow L^2(\nu_k)$ is well defined and a contraction, since (AU.10) gives the squared-norm inequality for every measurable function. It restricts to the identity on each polynomial space and commutes with multiplication by $c + iu$, remainder (AU.5), and every raw jet (AU.6). The comparison is therefore a map on the same quotient, not a declaration that the measures coincide.

Every moment of ν_k is finite, and the following integral evaluates it with positive integrands:

$$\nu_{k,j} := \int_{\mathbb{R}} u^j d\nu_k(u) = \begin{cases} \frac{2a_k j!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-d_k t}}{(\alpha + t)^{j+1}} dt, & j \text{ even,} \\ 0, & j \text{ odd.} \end{cases} \quad (\text{AU.12})$$

To prove it, repeated integration by parts gives $\int_0^\infty t^{B-1} e^{-yt} dt = (B-1)! y^{-B}$ for $y > 0$. Insert $y = d_k + u$ on the positive half-line. Tonelli applies because each integrand is nonnegative there. Integrating $u^j e^{-(\alpha+t)u}$ gives $j!(\alpha + t)^{-j-1}$ by the same integration by parts. Evenness supplies the factor two or the vanishing odd moment. The bound $\nu_{k,j} \leq 2a_k d_k^{-B} j! \alpha^{-j-1}$ for even j also proves finiteness directly. In particular $\nu_{k,0} > 0$ is retained, and generally differs from μ_h^k .

Let $H_N^\nu = (\nu_{k,i+j})_{0 \leq i,j \leq N}$. Its quadratic form is the ν_k integral of the polynomial's squared modulus. The weight is strictly positive everywhere, so a nonzero polynomial has positive norm. Thus $0 < H_N^\nu \preceq H_N$. Minimizing on the identical affine fibres in (AU.5) proves

$$0 < G_N^\nu \preceq G_N, \quad K_N \preceq K_N^\nu, \quad K_N^\nu = E_N(H_N^\nu)^{-1} E_N^*. \quad (\text{AU.13})$$

The inverse inequality follows also by conjugating H_N by $(H_N^\nu)^{-1/2}$ and inverting its positive eigenvalues.

Theorem 2.1 (Actual low-degree data control the higher quotient volume). *For the original quartet source and $k \geq 3$, the finite upper bound is*

$$0 \leq C_k \leq \log \det K_{2q-1}^\nu - \log \det K_q. \quad (\text{AU.14})$$

Its actual arithmetic inputs are the moments through degree $2q$ and the stated envelope constants. The comparator moments through $4q - 2$ are the explicit positive integrals (AU.12).

Proof. The first inequality follows by minimizing the same quotient norm in successively larger polynomial spaces. For the second use (AU.13) at $N = 2q - 1$ and (AU.9). Determinant is increasing on positive Hermitian forms: conjugate by the square root of the smaller one and multiply eigenvalues at least one. H_q uses precisely the original moments $0, \dots, 2q$, and all the matrices in the second term of (AU.14) have already specified finite algebraic maps from them. No higher actual moment enters. Equivalently, if $\mathfrak{D}_j^\nu = \det H_{j-1}^\nu$ and $\mathfrak{B}_j^\nu$ is the Gram determinant of the original ψ times $1, u, \dots, u^{j-1}$, the quotient-first coefficient matrix $[1, u, \dots, u^{q-1}, \psi, u\psi, \dots]$ is triangular with diagonal one. Orthogonal elimination of its relation block gives

$$V_{2q-1}^\nu = \mathfrak{D}_{2q}^\nu / \mathfrak{B}_q^\nu, \quad C_k \leq \log V_q + \log \mathfrak{B}_q^\nu - \log \mathfrak{D}_{2q}^\nu. \quad (\text{AU.15})$$

This proves the same upper bound in the exact source/relation determinants. Both the source mass and relation multiplication factors remain in their respective determinants. $\square$

The existing Gamma coefficient map also makes the cutoff in (AU.14) explicit. With $\lambda > 0$, $\alpha_\lambda = 2\lambda$, $c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda)$ and $C_\lambda = c_\lambda^k$, define the original monic Gamma polynomials by

$$\sum_{j \geq 0} b_j^{(\lambda)}(u) z^j / j! = (1 + z^2)^{-\lambda} e^{u \arctan z}, \quad c_{h,j} = \frac{\int b_j^{(\lambda)} w_h}{c_\lambda j! (\alpha_\lambda)_j}.$$

Multiplying the k generating functions and taking finite derivatives at zero gives

$$d_l = \sum_{j_1 + \dots + j_k = l} \prod_a (\alpha_\lambda)_{j_a} c_{h,j_a}, \quad \int b_l^{(k\lambda)} m_k = C_\lambda l! d_l. \quad (\text{AU.16})$$

All finite terms are integrable by the supplied exponential moments. Expand $u^{i+j} = \sum_{l \leq i+j} L_{ij,l} b_l^{(k\lambda)}(u)$ by monic subtraction. Then $(H_q)_{ij} = C_\lambda \sum_l l! L_{ij,l} d_l$. Thus the actual coefficient list $c_{h,0}, \dots, c_{h,2q}$ suffices for (AU.14). At zero degree it gives $C_\lambda c_{h,0}^k = \mu_h^k$, retaining the source mass.

2.3 An upper determinant depending only on the two Laplace values

Fix $0 < b < \pi/2$ and set $M_k = M_h(b)^k + M_h(-b)^k$, with the original $M_h(\pm b) = \int e^{\pm bu} w_h(u) du$. Positivity and convolution give $\int e^{b|u|} m_k(u) du \leq M_k$. The exponential series therefore gives $\int |u|^{2j} m_k(u) du \leq M_k (2j)! b^{-2j}$. For arbitrary $\tau_0, \dots, \tau_q > 0$ define

$$D_q(\tau) = M_k \left(\sum_{j=0}^q \tau_j \right) \text{diag} \left(\frac{(2j)!}{b^{2j} \tau_j} \right)_{j=0}^q. \quad (\text{AU.17})$$

For $f = \sum a_j u^j$, Cauchy–Schwarz applied to $\sqrt{\tau_j}$ and $a_j u^j / \sqrt{\tau_j}$ proves

$$|f(u)|^2 \leq \left(\sum \tau_j \right) \sum |a_j|^2 |u|^{2j} / \tau_j.$$

Integration proves $H_q \preceq D_q(\tau)$. Inverting and applying the same full jet map proves

$$\mathbf{K}_q \succeq \frac{1}{M_k \sum \tau_j} \mathbf{A}_q(\tau), \quad \mathbf{A}_q(\tau) = \mathbf{E}_q \operatorname{diag} \left(\frac{\tau_j b^{2j}}{(2j)!} \right) \mathbf{E}_q^* > 0. \quad (\text{AU.18})$$

Consequently an upper bound without any unevaluated actual polynomial moment is

$$C_k \leq q \log \left(M_k \sum_{j=0}^q \tau_j \right) + \log \det \mathbf{K}_{2q-1}^\nu - \log \det \mathbf{A}_q(\tau). \quad (\text{AU.19})$$

Every positive choice of τ gives a proved bound, and the infimum over such choices remains a bound. Multiplying all τ_j by the same positive number cancels between the two written determinant factors; it changes neither measure. No optimizer or numerical enclosure is assumed in (AU.19).

The lower determinant has the exact positive expansion

$$\det \mathbf{A}_q(\tau) = \sum_{\substack{I \subset \{0, \dots, q\} \\ |I|=q}} |\det(\mathbf{E}_q)_{[:,I]}|^2 \prod_{j \in I} \frac{\tau_j b^{2j}}{(2j)!}. \quad (\text{AU.20})$$

Expand each determinant multilinearly in columns in the product $\mathbf{E}_q D \mathbf{E}_q^*$: terms with repeated column indices vanish by alternation, and grouping permutations of the same subset gives (AU.20). Its subset $I = \{0, \dots, q-1\}$ is strictly positive by (AU.7). This expression retains full raw-jet minors and factorials without estimating an inverse by its individual entries.

2.4 The interval kernel and the actual comparison map

For $T > 0$, set

$$a_k(T) = a_k e^{-\alpha T} (d_k + T)^{-B}. \quad (\text{AU.21})$$

Restriction to $[-T, T]$ followed by the identity on polynomials maps the actual source to $L^2([-T, T], a_k(T) du)$ contractively. The same is true from $L^2(\nu_k)$, by the pointwise lower bound on that interval. Let $H_N^I(T)$ be the Lebesgue interval Gram. Then $a_k(T) H_N^I(T) \preceq H_N^\nu \preceq H_N$. Let P_j denote the Legendre polynomial specified by Rodrigues' formula, $P_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. Integration by parts j times proves its orthogonality to lower degrees; direct evaluation of the resulting beta integral gives its squared norm $2/(2j+1)$. Its highest coefficient is $(2j)!/(2^j (j!)^2)$. The endpoint boundary terms vanish because $(x^2 - 1)^j$ has zeros of order j at both endpoints. Applying these constants to $P_j(u/T)$ gives the full interval jet kernel

$$(\mathbf{K}_N^I(T))_{(a,b,d),(a',b',e)} = \sum_{j=0}^N \frac{2j+1}{2T^{1+d+e}} P_j^{(d)}(\zeta_{ab}/T) \overline{P_j^{(e)}(\zeta_{a'b'}/T)}. \quad (\text{AU.22})$$

This follows by writing the inverse Gram in the orthogonal Legendre basis and applying (AU.6); each derivative contributes its stated T^{-d} . In particular

$$\mathbf{K}_{2q-1} \preceq \mathbf{K}_{2q-1}^\nu \preceq a_k(T)^{-1} \mathbf{K}_{2q-1}^I(T). \quad (\text{AU.23})$$

Equations (AU.14) and (AU.19) remain valid if their $\log \det \mathbf{K}_{2q-1}^\nu$ is replaced by $\log \det \mathbf{K}_{2q-1}^I(T) - q \log a_k(T)$. The global envelope version is at least as strong as each such interval version by the proved positive-matrix inequality (AU.23).

The restriction map has the exact orthogonal decomposition $f \mapsto (f 1_{[-T,T]}, f 1_{\mathbb{R} \setminus [-T,T]})$ in the original weighted L^2 source; its two squared norms add to the original norm. The interval bound drops a nonnegative term in that equality. It does not assign the outside component to external absence.

2.5 The explicit scalar bound and its finite gap

The following explicit scalar estimate is the independent full calculation VC.1–43, composed here with the preceding original quotient maps. Put $R_0 = \sqrt{\delta^2 + \gamma^2}$ and define

$$\begin{aligned} D &= \max\{1, 2/b, 2R_0\}, \quad T = Dq, \quad r = kR_0/T \leq 1/2, \quad N = 2q - 1, \\ U_k(T) &= \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \quad A_N = \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \\ B_q(r) &= \left(\frac{1+r}{1-r}\right)^q, \quad \mathcal{C}_{k,q} = \frac{U_k(T) A_{2q-1}^2 B_q(r)^2}{Ta_k(T)}. \end{aligned} \quad (\text{AU.24})$$

All values are positive. In particular $q \geq k$ and $D \geq 2R_0$ give the stated bound on r . These choices bound constants; they do not change the original density.

Theorem 2.2 (A full finite Hermitian upper bound). *In the transported ordered quotient coordinates, and by the explicit congruence below also in the original S coordinates,*

$$\boxed{K_q \preceq K_{2q-1} \preceq \mathcal{C}_{k,q} K_q, \quad 0 \leq C_k \leq q \log \mathcal{C}_{k,q}.} \quad (\text{AU.25})$$

Proof. Let T_j be the coefficient matrix of the original map $\Psi(P) = P(c + iu)$ in degree j . Write $H_j^S, J_j^S, K_j^S, G_j^S$ for matrices in the original S monomials. Then

$$H_j^S = \mathsf{T}_j^* H_j \mathsf{T}_j, \quad J_j \mathsf{T}_j = \mathsf{T}_{q-1} J_j^S, \quad K_j = \mathsf{T}_{q-1} K_j^S \mathsf{T}_{q-1}^*.$$

The first identity is the source substitution, the second is monic division, and substitution in (AU.8) gives the third. Each $\det \mathsf{T}_j$ has modulus one, as proved in (AU.2). All source norms on S polynomials in this proof use H_j^S . The exact auxiliary coordinate map used in this proof is $\Psi_T P(x) = P(c + iTx)$, with inverse $f((S - c)/(iT))$. Its target measure is $Tm_k(Tx)dx$ and retains mass μ_h^k by substitution. The target monic relation is $\chi_T(x) = (iT)^{-q} \chi(c + iTx)$, so $\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q$. Monic division therefore proves $\text{rem}_{\chi_T} \Psi_T P = \Psi_T \text{rem}_{\chi} P$. At every original centre λ , the full order- d raw jet is $(\Psi_T P)^{(d)}((\lambda - c)/(iT)) = (iT)^d P^{(d)}(\lambda)$. The coefficient matrix has diagonal $(iT)^j$ at degree j ; its quotient determinant is $(iT)^{q(q-1)/2}$. Consequently the target quotient form is the original form transported by inverse congruence, and its determinant is $T^{-q(q-1)} V_j$ at every source degree j . This factor cancels in the central ratio, with all phases and masses still present in the map.

For a polynomial $f(x) = \sum f_j x^j$ put $\|f\|_{\text{coeff},1} = \sum |f_j|$. The Rodrigues formula above, expanded by the product rule, gives

$$P_j(x) = 2^{-j} \sum_{a=0}^j \binom{j}{a}^2 (x-1)^{j-a} (x+1)^a.$$

The coefficient norm of a product is at most the product of the two coefficient norms by convolution and the triangle inequality. Since $\|(x \pm 1)^a\|_{\text{coeff},1} = 2^a$, this proves $\|P_j\|_{\text{coeff},1} \leq \sum_a \binom{j}{a}^2 = \binom{2j}{j} \leq 4^j$. The middle equality is the coefficient of z^j in $(1+z)^j(1+z)^j$. In the orthogonal expansion $f = \sum_{j=0}^N f_j^L P_j$, Cauchy–Schwarz and the squared norm $2/(2j+1)$ give $|f_j^L| \leq \sqrt{(2j+1)/2} \|f\|_{L^2([-1,1])}$. Summing 4^j therefore proves

$$\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}, \quad \deg f \leq N. \quad (\text{AU.26})$$

List all roots $z_1, \dots, z_q$ of χ_T , with repetitions; each has modulus at most r . Define e_a, h_j by

$$\prod_{a=1}^q (1 - z_a t) = \sum_{a=0}^q (-1)^a e_a t^a, \quad \prod_{a=1}^q (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j.$$

The second identity is formal multiplication of geometric series. It gives $|e_a| \leq \binom{q}{a} r^a$ and $|h_j| \leq \binom{q+j-1}{j} r^j$: the latter counts exponent lists of total degree j by their $q-1$ separating bars. For $0 \leq s \leq q-1$, multiply those inverse series through order s . It proves that the quotient of x^{q+s} by χ_T is exactly $Q_s = \sum_{j=0}^s h_j x^{s-j}$ and its remainder is $x^{q+s} - \chi_T Q_s$, of degree below q . Removing the leading term of $\chi_T Q_s$ removes precisely one from its coefficient norm. All its remaining coefficients are the negative remainder. Thus

$$\begin{aligned} \|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} &\leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \\ &\leq ((1+r)/(1-r))^q = B_q(r). \end{aligned}$$

At degrees below q the remainder is the original monomial and has coefficient norm one, also bounded by $B_q(r)$. Linearity proves

$$\|\text{rem}_{\chi_T} f\|_{\text{coeff},1} \leq B_q(r) \|f\|_{\text{coeff},1}, \quad \deg f \leq 2q-1. \quad (\text{AU.27})$$

Every root was retained in this proof. Its smaller literal finite constant $\max\{1, (1+r)^q \sum_{j=0}^{q-1} \binom{q+j-1}{j} r^j - 1\}$ can also replace $B_q(r)$.

For v of degree at most $q-1$, each monomial is bounded by $\max\{1, |x|^{q-1}\}$. Using $\max(1, y) \leq 1+y$ for $y \geq 0$ and the proved Laplace moment bound gives

$$\int_{\mathbb{R}} |v(u/T)|^2 m_k(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2. \quad (\text{AU.28})$$

On the other hand, restriction to $[-T, T]$ and (AU.21) prove $\|P\|_{H_N^S}^2 \geq T a_k(T) \|\Psi_T P\|_{L^2([-1,1])}^2$. Take $f = \Psi_T P$ and $v = \text{rem}_{\chi_T} f$. Equations (AU.26)–(AU.28) and the exact division map now give

$$\|\text{rem}_{\chi} P\|_{H_{q-1}^S}^2 \leq C_{k,q} \|P\|_{H_N^S}^2. \quad (\text{AU.29})$$

The map estimated here is $s\pi_{\chi,N} : \mathcal{P}_N^S \rightarrow \mathcal{P}_{q-1}^S$, where $\pi_{\chi,N}$ is the polynomial quotient map and $s : \mathbb{C}[S]/(\chi) \rightarrow \mathcal{P}_{q-1}^S$ selects the unique remainder representative. Their exact identities are $\pi_{\chi,q-1} s = I$ and $\pi_{\chi,q-1} s \pi_{\chi,N} = \pi_{\chi,N}$. The composite $s\pi_{\chi,N}$ equals the identity on polynomials of degree below q .

Apply (AU.29) to the least lift $R_N^S x$ from the original matrices. Its remainder lifts the same x already at degree $q-1$, hence $x^* G_q^S x \leq x^* G_{q-1}^S x \leq C_{k,q} x^* G_N^S x$. Conversely degree inclusion gives $G_N^S \preceq G_q^S$. Conjugating by $(G_q^S)^{-1/2}$, bounding its positive eigenvalues and then inverting proves (AU.25), including the determinant exponent q , first for K_j^S and then by the displayed congruence for K_j . It also proves $C_{k,q} \geq 1$. $\square$

Retain the fixed constants

$$X = \max\{1, M_h(b), M_h(-b)\}, \quad K = \max\{1, \vartheta_h^{-1}\}, \quad C_{\text{env}} = \max\{1, 2/(3c_h D)\}.$$

Because $bT \geq 2q$, the factorial ratio in (AU.24) is at most one. Also $M_k = \int 2 \cosh(bu) m_k(u) du \geq 2\mu_h^k$. It follows that $U_k(T) \leq 3X^k$. The displayed A_N satisfies $A_{2q-1}^2 \leq (2q/9)256^q$. On $[0, 1/2]$ the derivative of $\log((1+r)/(1-r))$ is $2/(1-r^2) \leq 8/3$, so

$$\log B_q(r)^2 \leq \frac{16}{3} q r = \frac{16R_0}{3D} k. \quad (\text{AU.30})$$

Use also $\vartheta_h^{-(k-3)} \leq K^k$, $1+Dq+k-3 \leq 1+(D+1)q$, and $e^{-3\alpha} \leq 1$. Substitution in (AU.24)–(AU.25) proves the finite scalar bound

$$\begin{aligned} 0 \leq C_k \leq \mathcal{U}_k &:= (\alpha D + \log 256) q^2 \\ &+ \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) k q \\ &+ Bq \log(1 + (D+1)q) + q \log C_{\text{env}}. \end{aligned} \quad (\text{AU.31})$$

All of its constants are independent of k and q for fixed h, b . The exact finite bounds (AU.14), (AU.19), and $q \log C_{k,q}$ can be retained together by taking their minimum; each independently bounds the same C_k with the proved maps.

The signed endpoint calculation now improves this earlier central estimate on the same original source. Retain its definitions $\mathcal{B}_{h,k}^\Gamma$, $\mathcal{J}_{h,k}$, the recursively refined upper endpoint $\mathcal{B}_{h,k}^{\text{hi}}$, and the arithmetic endpoint terms $E_0 = \log(V_{q-1}/V_q)$, $E_1 = \log(V_{2q-1}/V_{2q})$ from (FV.M1)–(FV.M18), Section 1.1.6. The exact coefficient map Ψ in (AU.2) carries its S source to this u source. Its quotient coefficient map τ_{q-1} sends $[P(S)]$ to $[P(c + iu)]$; because $\Psi(\chi Q) = i^q \psi \Psi(Q)$, it has the exact remainder identity $J_N^u \tau_N = \tau_{q-1} J_N^S$. Thus $K_N^u = \tau_{q-1} K_N^S \tau_{q-1}^*$ and $V_N^u = |\det \tau_{q-1}|^{-2} V_N^S = V_N^S$. The identity is valid at both densities of the interpolation; their literal masses, the factor i^q in relation multiplication and every raw-derivative factor i^d are retained. Define

$$\mathcal{U}_k^{\text{signed}} = \frac{\mathcal{B}_{h,k}^{\text{hi}} - E_0 - E_1}{2}. \quad (\text{AU.31a})$$

Equations (AU.9) and (FV.M11) now prove $0 \leq C_k \leq \mathcal{U}_k^{\text{signed}}$ by substitution, with no endpoint term discarded. Together with the existing proved bounds this gives

$$0 \leq C_k \leq \widehat{\mathcal{U}}_k := \min \left\{ \log \det \mathcal{K}_{2q-1}^\nu - \log \det \mathcal{K}_q, \quad q \log C_{k,q}, \quad \mathcal{U}_k, \quad \mathcal{U}_k^{\text{signed}}, \right. \\ \left. q \log \left(M_k \sum_{j=0}^q \tau_j \right) + \log \det \mathcal{K}_{2q-1}^\nu - \log \det \mathcal{A}_q(\tau) \right\}, \quad \tau_j > 0. \quad (\text{AU.31b})$$

Each entry is an upper bound for the same C_k , respectively by (AU.14), (AU.25), (AU.31), (FV.M11), and (AU.19). Their minimum is therefore an upper bound; (AU.23) still gives each interval-kernel variant with its full T powers. The new signed entry includes the actual-moment full-angle refinement and its nonlinear transport in (FV.M15)–(FV.M18). It uses the actual original source moments through degree $4q$, the relative eigenvalues and the actual projection overlap in (FV.M7). The lower-degree arithmetic cutoff of (AU.14) is unchanged for its own entry; it is not asserted for the new signed entry.

The complete central lower estimate in the balanced-window/factor-four source is $C_k \geq \mathcal{L}_k = 2(q-1)(\log k + \log(\delta/(2C_h^{\text{bal}})))$. The current explicitly computed gap is the literal function

$$\mathcal{U}_k - \mathcal{L}_k = (\alpha D + \log 256)q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ + Bq \log(1 + (D+1)q) + q \log C_{\text{env}} \\ - 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (\text{AU.32})$$

The propagated upper bound gives the additional actual finite gap $0 \leq \widehat{\mathcal{U}}_k - \mathcal{L}_k \leq \mathcal{U}_k - \mathcal{L}_k$: the first inequality follows from $\mathcal{L}_k \leq C_k \leq \widehat{\mathcal{U}}_k$, and the second from the minimum in (AU.31b). The function in (AU.32) still concerns the two displayed scalar upper and lower estimates. It is not an asserted exact value of the unknown C_k . Since $q = \ell(k+1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. Dividing every term of (AU.31)–(AU.32) by q^2 proves

$$\lim_{k \rightarrow \infty} \frac{\mathcal{U}_k - \mathcal{L}_k}{q_k^2} = \alpha D + \log 256 > 0, \quad \limsup_{k \rightarrow \infty} C_k/q_k^2 \leq \alpha D + \log 256. \quad (\text{AU.33})$$

Dividing the upper estimate by $q \log k$ gives the exact leading factor $(\alpha D + \log 256)q/\log k$. It grows as $(\alpha D + \log 256)k^2/\log k$ for $m = 1$ and as $(\alpha D + \log 256)(m-1)k^3/\log k$ for $m > 1$. The division step itself has only the linear-in- k logarithmic cost (AU.30). The two quadratic contributions in this scalar proof come from the full coefficient estimate (AU.26) and the least density over the interval in (AU.21). The positive matrices in (AU.14), (AU.19), and (AU.22) retain the directional information discarded in those scalar estimates, with their exact source and quotient maps already proved. No upper asymptotic for those sharper determinants is inserted.

2.6 A full-jet quadratic benchmark and its exact relation family

Here the auxiliary source is explicitly Lebesgue measure on $[-1, 1]$ and the auxiliary relation is u^q . Its mass is two. These data are inserted into the same sequence (AU.5), kernel (AU.8), and central ratio (AU.9). No assertion that they are an arithmetic density is made. Write D_d for its d by d monomial determinant, $D_0 = 1$, and $B_j(q)$ for the j by j determinant of the relation measure $u^{2q} du$ on that interval. The quotient-first triangular determinant calculation proving (AU.15) gives exactly

$$\frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} = \frac{D_{q+1}B_q(q)}{B_1(q)D_{2q}}, \quad B_1(q) = \frac{2}{2q+1}. \quad (\text{AU.40})$$

The orthogonal Legendre calculation gives

$$D_d = \prod_{j=0}^{d-1} \frac{2}{2j+1} \left(\frac{2^j}{\binom{2j}{j}} \right)^2. \quad (\text{AU.41})$$

For $q = 2s$, parity splits the relation Gram into two s by s blocks, and

$$B_{2s}(2s) = \mathcal{C}_s(2s+1/2)\mathcal{C}_s(2s+3/2), \quad \mathcal{C}_s(x) = \frac{\prod_{j=0}^{s-1} (j!)^2}{\prod_{a,b=0}^{s-1} (x+a+b)}. \quad (\text{AU.42})$$

Indeed the even block has entries $1/(2s+a+b+1/2)$ and the odd block entries $1/(2s+a+b+3/2)$. The Cauchy determinant formula used here has numerator $\prod_{a<b} (b-a)^2 = \prod_j (j!)^2$. It follows by multiplying the determinant by all its denominators: the numerator is alternating separately in its two lists of variables, so contains their two Vandermonde factors. Its degree is their total degree, and comparing one residue inductively gives constant one. Substitution gives (AU.42) with no absolute-value or sign ambiguity.

Theorem 2.3 (An explicit quadratic scale in the full-origin jet fibre). *For every even integer $q \geq 2$,*

$$\left| \log \frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} - \kappa_* q^2 \right| \leq 12q \log(8q), \quad \kappa_* := \frac{9}{2} \log 3 - 6 \log 2 = \frac{3}{2} \log(27/16) > 0. \quad (\text{AU.43})$$

Proof. The elementary binomial bounds $4^j/(2j+1) \leq \binom{2j}{j} \leq 4^j$ hold because the central coefficient is the largest of the $2j+1$ coefficients summing to 4^j . Therefore (AU.41) gives

$$\log D_d = -d(d-1) \log 2 + E_d, \quad |E_d| \leq d \log(4d) \quad (d \geq 1). \quad (\text{AU.44})$$

At $s \geq 1$, expand the numerator of $\mathcal{C}_s(2s)$ as $2 \sum_{i=1}^{s-1} (s-i) \log i$. The function $f(x) = (1-x) \log x$ is increasing on $(0, 1]$, has integral $-3/4$, and is nonpositive. Its right Riemann sum differs from its integral by a number between zero and $(\log s+1)/s$: the difference is bounded above by $-\int_0^{1/s} f(x) dx \leq (\log s+1)/s$, using $-f(x) \leq -\log x$. It follows that this numerator is $s^2 \log s - 3s^2/2$ plus an error of absolute value at most $s \log s + 2s$.

The denominator's logarithm is $s^2 \log s + \sum_{a,b=0}^{s-1} \log(2+a/s+b/s)$. Both partial derivatives of the latter function lie in $[0, 1/2]$. Its left-square Riemann sum differs from its integral by at most $1/s$, hence the sum by at most s . Two integrations give

$$\int_0^1 \int_0^1 \log(2+x+y) dx dy = 18 \log 2 - 9 \log 3 - 3/2.$$

For example use the second antiderivative $F(t) = t^2 \log t/2 - 3t^2/4$ and calculate $F(4) - 2F(3) + F(2)$. Thus

$$|\log \mathcal{C}_s(2s) - 9s^2 \log(3/4)| \leq s \log s + 3s.$$

For $c > 0$, the change to $\mathcal{C}_s(2s + c)$ has logarithm $-\sum_{a,b} \log(1 + c/(2s + a + b))$, between $-cs/2$ and zero. Using $c = 1/2, 3/2$ in (AU.42) proves

$$\left| \log B_q(q) - \frac{9}{2}q^2 \log(3/4) \right| \leq q \log(q/2) + \frac{7}{2}q. \quad (\text{AU.45})$$

Insert (AU.44)–(AU.45) into (AU.40). The quadratic coefficient is $3 \log 2 + (9/2) \log(3/4) = \kappa_*$. The remaining absolute error is at most

$$3q \log 2 + (q + 1) \log(4(q + 1)) + 2q \log(8q) \\ + q \log(q/2) + \frac{7}{2}q + \log((2q + 1)/2).$$

For $q \geq 2$, use $q + 1 \leq 3q/2$, $4(q + 1) \leq 8q$, $\log(q/2) \leq \log(8q)$ and $\log((2q + 1)/2) \leq \log(8q)$. The two constant multiples of q are bounded together by $3q \log(8q)$, since $3 \log(16) > 3 \log 2 + 7/2$. Their sum is then at most $8q \log(8q)$, hence certainly the displayed $12q \log(8q)$. This proves (AU.43) for every indicated finite q , not only along a limit. $\square$

There is an exact algebraic relation of this benchmark's quotient to every original grid quotient. Define

$$\psi_t(u) = \prod_{a,b} (u - t\zeta_{ab})^\ell, \quad \mathcal{A} = \mathbb{C}[t, u]/(\psi_t(u)). \quad (\text{AU.46})$$

Monic division over $\mathbb{C}[t]$ proves that $1, u, \dots, u^{q-1}$ form a free $\mathbb{C}[t]$ basis: division gives spanning and a nonzero multiple of a monic degree- q polynomial cannot have smaller degree. Every fibre has dimension q . The fibre at $t = 1$ is the original transported quotient; at $t = 0$ it is $\mathbb{C}[u]/u^q$. For $t \neq 0$,

$$\mathcal{A}_t \xrightarrow{\sim} \mathcal{A}_1, \quad [f(u)] \mapsto [f(tu)], \quad [g(u)] \mapsto [g(u/t)] \quad (\text{AU.47})$$

are inverse maps, since $\psi_t(tu) = t^q \psi_1(u)$. Their full raw jet equation is $(d/du)^d f(tu)|_{u=\zeta_{ab}} = t^d f^{(d)}(t\zeta_{ab})$, so the coefficient d in multiplication's raw-jet Jordan blocks and each order ℓ remain for $t \neq 0$. At $t = 0$ monic division gives the complete merged order q , rather than an assertion that separate evaluation rows stay invertible at coincident points.

For any fixed positive polynomial source H_N , remainder matrices $J_N(t)$ are polynomial in t by the monic subtraction algorithm. They contain I_q in their first q columns. Thus $K_N(t) = J_N(t)H_N^{-1}J_N(t)^*$ is positive for every real t , including zero, and the exact central function obeys

$$C(t) = \log \det K_{2q-1}(t) - \log \det K_q(t), \quad C'(t) = \text{Tr}(K_{2q-1}^{-1}K'_{2q-1} - K_q^{-1}K'_q), \quad (\text{AU.48})$$

where $K'_N = J'_N H_N^{-1} J_N^* + J_N H_N^{-1} (J'_N)^*$. The determinant derivative follows by its multilinear expansion. This gives both the free family and the exact finite source-dependent map along it. The estimate (AU.43) belongs to its stated Lebesgue special fibre. Equations (AU.46)–(AU.48) do not transfer that estimate to $t = 1$ without the displayed change in the full kernel.

2.7 Source representatives and support

The source polynomial maps throughout are the original $\mathcal{V}_{h,k}P = P(D_1 + \dots + D_k)F_h^{\otimes k}$, $D = -x\partial_x$, $\mathcal{M}F_h = g/h$. The arithmetic inclusion is the original $\eta[P] = [g/h]_h^{\otimes k} P(\sum s_i)$, after the explicit chart (AU.2). Each local Taylor constant of g/h at a selected complete zero is nonzero. Recursively solving multiplication of truncated Taylor series therefore proves this factor is a unit. It does not alter the original cyclic kernel (AU.4). The comparison to ν_k changes the norm used in an estimate, while its identity polynomial map leaves η , every raw jet, and the original relations unchanged.

For two polynomial representatives with the same quotient, monic division gives their difference χQ . In the tensor variables the original annihilator identity supplies $\chi(\sum s_i) = \sum_i h(s_i)Q_i(s)$,

with Q_i obtained by successive fixed-order monic divisions. The final remainder is zero because the monomial tensor basis is independent and the left side annihilates the sum. If $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$ and $h(D)F_h = \Theta\phi_0$, the actual primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D) Q(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{AU.49})$$

The tensor differential at position i contributes the second $(-1)^{i-1}$, cancelling the displayed one and giving exactly $\mathcal{V}_{h,k}(\chi Q)$. Thus every quotient comparison retains the same cohomology class with an explicit original primitive. On the supported carrier a fibre map sends (λ, v) to (λ, Lv) and sends external τ to τ . The killed relation has image $(\lambda, 0) = e_\lambda$. No comparison identifies that supported zero with external absence.

2.8 Verification and provenance

The complete arithmetic source (including the proof of its envelope), the exact endpoint-window source, the full PR24 note and scope correction/review, and the owner's complete balanced-window/factor-four proof were read for this calculation. Its new work is (AU.11)–(AU.23), the scalar estimate (AU.24)–(AU.33), and the full calculation (AU.40)–(AU.48). The constants $c_h, \vartheta_h, M_h(\pm b)$ retain their supplied definitions; no numerical enclosure of actual quartet data is claimed. The accompanying finite rational checker verifies the displayed maps and determinant formulas on declared auxiliary positive sources. It does not certify the analytic source by sampled values and makes no Lean claim.

3 Exact arithmetic restriction certificates and projection products

This article joins the original arithmetic lower-envelope comparison to the canonical restriction maps. It proves a positive arithmetic return gap at every finite degree, carries entry errors through the actual quotient, and identifies all loss traces with observations of two original source projections. The balanced product gives a complementary lower estimate on every inverse spectral power in both endpoint windows. The complete arithmetic envelope and balanced monic-norm proofs are retained as explicit source dependencies; their constants appear below. Every finite matrix and restriction consequence used here is proved.

3.1 The original source, quotient, and canonical lifts

Retain $g = 2\xi$, the monic packet h of selected zeros with their complete orders, $v_h = g/h$, and

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad c = k/2. \quad (\text{ERJ.1})$$

The original pairing, conjugate-linear in its first entry, is $\langle P, Q \rangle = \int \overline{P(c + iu)} Q(c + iu) m_k(u) du$. Its constant norm is μ_h^k . The arithmetic construction supplies positivity almost everywhere and all polynomial moments. A nonzero polynomial has finitely many zeros on this line, so its norm is positive. Let $S = s_1 + \dots + s_k$, let the monic generator of $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$ be χ , and put $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi \geq 1$, $A = M_S$. The ordered quotient coordinates are $1, S, \dots, S^{q-1}$. For $N \geq q - 1$ write $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{D}_N = \chi \mathcal{P}_{N-q}$, with negative degree spaces zero. Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0. \quad (\text{ERJ.2})$$

In monomial coordinates let H_N be the source Gram. In the original monic orthogonal basis $p_0, \dots, p_N$, put $O_N = \text{diag}(\omega_0, \dots, \omega_N)$ and $B_N = [b_0, \dots, b_N]$, $b_n = [p_n]_\chi$. All p_n stay monic;

$\omega_0 = \mu_h^k$. The monic triangular coefficient change has determinant one and gives the equal constructions

$$\begin{aligned} K_N &= J_N H_N^{-1} J_N^* = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \\ R_N &= O_N^{-1} B_N^* G_N : (E, G_N) \longrightarrow (\mathcal{P}_N, O_N), \quad K_N = \sum_{n=0}^N \frac{b_n b_n^*}{\omega_n}. \end{aligned} \quad (\text{ERJ.3})$$

Here and below a formula with O_N uses monic source coordinates; its monomial version conjugates by that explicit coefficient change. Surjectivity of B_N proves $K_N > 0$. Multiplication gives $B_N R_N = I$ and $R_N^* O_N R_N = G_N$. For $z \in \ker B_N$, $z^* O_N R_N = z^* B_N^* G_N = 0$. Thus every lift $R_N x + z$ has squared norm $x^* G_N x + z^* O_N z$. This proves the unique minimum property, $\text{im } R_N = \mathcal{D}_N^\perp$, and, for $i \leq j$, $G_j \preceq G_i$ and $K_i \preceq K_j$.

The actual analytic representative is $\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}$, where $D = -x\partial_x$, $\mathcal{M}F_h = v_h$, and $h(D)F_h = \Theta\phi_0$. Its norm is the pairing in (ERJ.1). Its full-jet map is $J^{(k)}\mathcal{V} = \eta J_N$, where $\eta[P] = [v_h]_h^{\otimes k} P(A_k)1$. Each local factor of $[v_h]_h$ has nonzero constant coefficient, so its inverse is the finite geometric expansion in its nilpotent part. The kernel of $P \mapsto P(A_k)1$ is exactly (χ) by its definition. These two facts prove that η is injective, retaining every local derivative and arithmetic unit.

3.2 Return, action, and the entire original relation correction

Let $L_{i,j} : \mathcal{P}_i \hookrightarrow \mathcal{P}_j$ pad monic coefficients, and let $P_{i,j}$ be the orthogonal projection retaining their first $i+1$ entries. Then $B_j L_{i,j} = B_i$ and $L_{i,j}^* O_j L_{i,j} = O_i$. The original identity transport $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$, $x \mapsto x$, is a contraction and intertwines A . Its metric adjoint is

$$T_{i,j} = K_i G_j : (E, G_j) \longrightarrow (E, G_i), \quad L_{i,j} R_i T_{i,j} = P_{i,j} R_j, \quad T_{i,j} T_{j,l} = T_{i,l}. \quad (\text{ERJ.4})$$

Indeed $x^* G_j y = x^* G_i T_{i,j} y$. The first $i+1$ coordinates of R_j are $O_i^{-1} B_i^* G_j$, exactly those of $R_i T_{i,j}$. The composition identity is $K_i G_j K_j G_l = K_i G_l$. For endomorphism types put $U_{i,j} = T_{i,j} I_{i,j}$ on (E, G_i) and $\widehat{U}_{i,j} = I_{i,j} T_{i,j}$ on (E, G_j) . Their coordinate matrices are $K_i G_j$, and the original invertible identity transport satisfies $I_{i,j} U_{i,j} = \widehat{U}_{i,j} I_{i,j}$. An endomorphism power or eigenvalue of T below means these specified similar endomorphisms, with all metrics retained. As an underlying endomorphism of E , $T = T_{i,j}$ is self-adjoint and positive in both displayed metrics: $G_i T = G_j$ and $G_j T = G_j K_i G_j$. It obeys $0 < T \preceq I$ in each metric. Its eigenvalues g_a lie in $(0, 1]$. The squared singular values of $I_{i,j}$, of its adjoint in the reverse Hilbert types, and of the restricted canonical lift in (ERJ.4) are these same g_a . Consequently

$$H_{i,j}^{\text{loss}} = I - T_{i,j}, \quad \mathcal{L}_{i,j} := \log(V_i/V_j) = -\log \det T_{i,j} = -\sum_{a=1}^q \log g_a. \quad (\text{ERJ.5})$$

Write $W_N = A^* G_N + G_N A - k G_N$, $H_N = K_N W_N$, and $V_N = A K_N + K_N A^* - k K_N$. Direct expansion proves

$$AT - TA = H_i T - T H_j = V_i G_j - K_i W_j. \quad (\text{ERJ.6})$$

The two $K_i A^* G_j$ terms and the two $k K_i G_j$ terms cancel. If $\epsilon_N = \|H_N\|_{G_N}$, contraction of the adjoint gives $\|AT - TA\|_{(E, G_j) \rightarrow (E, G_i)} \leq \epsilon_i + \epsilon_j$. Thus the original action accompanies the return through its exact commutator, including on any chosen return eigenspace. The representative change is the entire map

$$\begin{aligned} \Delta_{i,j} &= L_{i,j} R_i - R_j = L_{i,j} R_i H_{i,j}^{\text{loss}} - (I - P_{i,j}) R_j, \\ B_j \Delta_{i,j} &= 0, \quad \text{im } \Delta_{i,j} \subseteq \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \Delta_{i,j}^* O_j \Delta_{i,j} = G_i - G_j. \end{aligned} \quad (\text{ERJ.7})$$

Both terms in the difference have remainder $H_{i,j}^{\text{loss}}$. Orthogonality to $\mathcal{D}_i$ follows for both complete lifts. Since R_j is orthogonal to $\mathcal{D}_j$, the equality $L_{i,j} R_i = R_j + \Delta_{i,j}$ is an orthogonal sum and proves the Gram.

For the explicit theta primitive keep $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees zero and one, with F_h of degree one and ϕ_0 of degree zero. The original compatible chain action D commutes with this differential. Successive monic division gives $\chi(\sum s_a) = \sum_a h(s_a)Q_a(s)$ in the tensor ideal. For any relation $\chi\kappa$ the cochain

$$\sum_{a=1}^k (-1)^{a-1} Q_a(D) \kappa(\sum_b D_b) (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}) \quad (\text{ERJ.8})$$

has differential $\mathcal{V}(\chi\kappa)$: applying the tensor differential gives a second factor $(-1)^{a-1}$ and $h(D_a)F_h = \Theta\phi_0$; all other terms have zero differential. Apply this fixed linear division primitive to every column of (ERJ.7). On an active split fibre the maps are $x^\bullet \mapsto x^\bullet$ and $x^\bullet \mapsto (Tx)^\bullet$, with $\tau \mapsto \tau$. They are $G(\mathbb{C})$ -linear by the original fibre operations and linearity of the displayed maps. The relation image in the internal quotient is the supported zero $(\ell_j, 0)$; the external τ is retained. The reverse maps compose by (ERJ.4), while their action comparison is (ERJ.6). No relation, support label, or primitive is deleted.

3.3 The exact source projection and principal-angle morphism

Use the common original Hilbert space $\mathcal{P}_j$ and its isometries $\ell_i = L_{i,j}R_i : (E, G_i) \rightarrow \mathcal{P}_j$ and $\ell_j = R_j : (E, G_j) \rightarrow \mathcal{P}_j$. Let Π_i, Π_j be the orthogonal endomorphism projections onto these two images. Because $R_j \perp \mathcal{D}_j \supseteq \mathcal{D}_i$, its lower-degree projection lies in $\ell_i E$. Also $\ell_i x - \ell_j x \in \mathcal{D}_j$. Hence

$$\Pi_i \ell_j = P_{i,j} \ell_j = \ell_i T, \quad \Pi_j \ell_i = \ell_j, \quad (\Pi_i \Pi_j)|_{\ell_i E} = \ell_i T \ell_i^{-1}, \quad (\Pi_j \Pi_i)|_{\ell_j E} = \ell_j T \ell_j^{-1}. \quad (\text{ERJ.9})$$

These are similarities on the indicated full images. The range of $\Pi_i \Pi_j$ is $\ell_i E$ because its restriction there is invertible. Its kernel is $\ker \Pi_j$: the reverse inclusion follows from invertibility of $\Pi_i : \ell_j E \rightarrow \ell_i E$. This gives its direct sum into that image and kernel, with exactly q nonzero eigenvalues g_a and $j+1-q$ zero eigenvalues. Reversing the pair proves the corresponding assertion for $\Pi_j \Pi_i$.

Choose a G_i -orthonormal eigenbasis x_a of T . Put $u_a = \ell_i x_a$ and $v_a = \ell_j x_a / \sqrt{g_a}$. Since $G_j = G_i T$, both families are orthonormal and $\langle u_a, v_b \rangle = \delta_{ab} \sqrt{g_a}$ by (ERJ.9). If $g_a = 1$ then $u_a = v_a$. Otherwise $w_a = (v_a - \sqrt{g_a} u_a) / \sqrt{1 - g_a}$ makes mutually orthonormal pairs (u_a, w_a) . On each such plane,

$$\Pi_i = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \Pi_j = \begin{pmatrix} g_a & \sqrt{g_a(1-g_a)} \\ \sqrt{g_a(1-g_a)} & 1-g_a \end{pmatrix}. \quad (\text{ERJ.10})$$

Their difference has trace zero and square $(1 - g_a)I_2$. Both vanish on the common orthogonal complement, and the shared directions with $g_a = 1$ contribute zero. Therefore, for every real $s > 0$ and in particular each integer $m \geq 1$,

$$\text{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2s} = 2 \sum_a (1 - g_a)^s, \quad \text{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2m} = 2 \text{Tr}_E ((H_{i,j}^{\text{loss}})^m). \quad (\text{ERJ.11})$$

Functional powers here use the proved positive spectral decomposition. The restricted map $\mathcal{R} = \Pi_i|_{\ell_j E}$ is the bijection $\ell_i T_{i,j} \ell_j^{-1} : \ell_j E \rightarrow \ell_i E$. For every $s > 0$ the exact inverse singular-value-power observation is

$$\sum_a \sigma_a(\mathcal{R}^{-1})^{2s} = \sum_a \sigma_a(T_{i,j}^{-1})^{2s} = \sum_a \sigma_a(I_{i,j}^{-1})^{2s} = \text{Tr} U_{i,j}^{-s} = \sum_a g_a^{-s}. \quad (\text{ERJ.11a})$$

This follows from $I_{i,j}^\dagger I_{i,j} = U_{i,j}$ and the equal singular values of a map and its adjoint; inversion reciprocates the singular values. These are Schatten observations of the original typed maps, including when $0 < 2s < 1$, where no norm triangle inequality is asserted. Only the bijection between the canonical images is inverted; the full projection products retain their ambient zeros. For a return eigenvector x of eigenvalue g the original lower portion satisfies

$$\frac{\|P_{i,j} R_j x\|_{\mathcal{O}_j}^2}{\|R_j x\|_{\mathcal{O}_j}^2} = g, \quad \frac{\|(I - P_{i,j}) R_j x\|_{\mathcal{O}_j}^2}{\|R_j x\|_{\mathcal{O}_j}^2} = 1 - g. \quad (\text{ERJ.12})$$

Indeed its first squared norm is $x^*G_jK_iG_jx = gx^*G_jx$; orthogonal decomposition proves the second. Its old quotient coordinate is gx . Applying the invertible T^{-1} recovers x ; the complete representative difference is exactly (ERJ.7).

3.4 The arithmetic remainder bound at every finite degree

For the actual quartet $1/2 \pm \delta \pm i\gamma$ of common order $m \geq 1, \delta, \gamma > 0$, retain

$$\ell = 1 + k(m-1), \quad q = \ell(k+1)^2, \quad \chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^\ell. \quad (\text{ERJ.13})$$

The top nonzero power of the sum of k local length- m nilpotents is $k(m-1)$, with coefficient $(k(m-1))!/((m-1)!)^k$ on the product of their top powers; its next power vanishes by total degree. Each displayed grid sum occurs and the distinct pairs have distinct real and imaginary parts. This proves the complete order and degree in (ERJ.13) by taking the least common multiple of local annihilators.

The complete arithmetic-envelope source proves, for $k \geq 3$,

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h, \quad \vartheta_h = \int_{-1}^1 w_h > 0. \quad (\text{ERJ.14})$$

Here $c_h > 0$ is its original local-mass convolution constant. Fix $0 < b < \pi/2$, $M_h(\pm b) = \int e^{\pm bt} w_h(t) dt$ and $M_k = M_h(b)^k + M_h(-b)^k$. Positivity and convolution give $\int e^{b|u|} m_k \leq M_k$, and hence $\int |u|^{2d} m_k \leq M_k (2d)! b^{-2d}$. Set $R_0 = \sqrt{\delta^2 + \gamma^2}$, $D = \max(1, 2/b, 2R_0)$, $T = Dq$, $r_0 = kR_0/T \leq 1/2$, and retain

$$\begin{aligned} a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(k-3+T)} (k-2+T)^{-B}, \\ U_k(T) &= \mu_h^k + M_k \frac{(2q-2)!}{(bT)^{2q-2}}, \quad A_N = \sqrt{(2N+1)/2} \frac{4^{N+1}-1}{3}, \\ B_q(r_0) &= \left(\frac{1+r_0}{1-r_0} \right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r_0)^2}{T a_k(T)}. \end{aligned} \quad (\text{ERJ.15})$$

The coordinate map $\Psi_T P(x) = P(c + iTx)$ has inverse $f \mapsto f((S-c)/(iT))$ and target measure $T m_k(Tx) dx$, whose mass remains μ_h^k . Put $\chi_T = (iT)^{-q} \chi(c + iTx)$. Then $\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q$, so remainder commutes with this map. Every original raw derivative of order d is multiplied by $(iT)^d$. The quotient coefficient determinant is $(iT)^{q(q-1)/2}$, and its transported form determinant is $T^{-q(q-1)} V_N$; this identical factor cancels in a volume ratio.

List all q roots z_a of χ_T with repetition. Their moduli are at most r_0 . Write $E(t) = \prod_a (1 - z_a t)$ and $E(t)^{-1} = \sum_{l \geq 0} h_l t^l$. For every integer $s \geq 0$, multiplication of these formal inverse series through order s proves that $Q_s(x) = \sum_{l=0}^s h_l x^{s-l}$ is the quotient of x^{q+s} by χ_T , and that $x^{q+s} - \chi_T Q_s$ has degree below q . Convolution gives $\|\chi_T\|_1 \leq (1+r_0)^q$ and $|h_l| \leq \binom{q+l-1}{l} r_0^l$, where $\|\cdot\|_1$ is the coefficient absolute sum. The sole coefficient of $\chi_T Q_s$ at degree at least q is its leading coefficient one. Its remaining coefficients are the negative remainder. Therefore

$$\|\text{rem}_{\chi_T} x^{q+s}\|_1 \leq (1+r_0)^q \sum_{l=0}^s \binom{q+l-1}{l} r_0^l - 1 \leq B_q(r_0). \quad (\text{ERJ.16})$$

The last bound sums the q geometric series. At degrees below q the remainder has norm one. Linearity proves $\|\text{rem}_{\chi_T} f\|_1 \leq B_q(r_0) \|f\|_1$ for every polynomial f , without a degree restriction on s .

Rodrigues' formula for Legendre polynomials gives orthogonality, squared norm $2/(2n+1)$, and coefficient norm at most $\binom{2n}{n} \leq 4^n$. For the norm, integrate by parts n times; the endpoint terms

vanish because $(x^2 - 1)^n$ has order n there, and the remaining beta integral gives $2/(2n + 1)$. For the coefficient bound use $P_n(x) = 2^{-n} \sum_{a=0}^n \binom{n}{a}^2 (x - 1)^{n-a} (x + 1)^a$, then sum the coefficient absolute values. Cauchy-Schwarz on each orthogonal coefficient and summation of 4^n yield $\|f\|_1 \leq A_N \|f\|_{L^2[-1,1]}$ for $\deg f \leq N$. For $\deg v \leq q - 1$, each monomial at u/T has modulus at most $\max(1, |u/T|^{q-1})$, so $\int |v(u/T)|^2 m_k \leq U_k(T) \|v\|_1^2$. Finally (ERJ.14) restricted to $[-T, T]$ gives $\|P\|^2 \geq T a_k(T) \|\Psi_T P\|_{L^2[-1,1]}^2$. Composing these actual maps proves, for every $N \geq q - 1$,

$$\|\text{rem}_\chi P\|^2 \leq C_N \|P\|^2, \quad G_{q-1} \preceq C_N G_N, \quad C_j^{-1} K_j \preceq K_i \preceq K_j \quad (q - 1 \leq i \leq j). \quad (\text{ERJ.17})$$

Apply the first inequality to the least lift $R_N x$ to obtain the second. Then use $G_j \preceq G_i \preceq G_{q-1}$ and inversion of positive forms. Taking $i = j$ proves $C_j \geq 1$. Thus the two original endpoint pairs $(q - 1, 2q - 1)$ and $(q, 2q)$ have the explicit positive gaps C_{2q-1}^{-1} and C_{2q}^{-1} .

3.5 Finite comparison matrices and the complete trace error

All following comparisons use the original quotient coordinates. For positive source brackets $H_N^- \preceq H_N \preceq H_N^+$, inversion and congruence give

$$L_N = J_N (H_N^+)^{-1} J_N^* \preceq K_N \preceq U_N = J_N (H_N^-)^{-1} J_N^*. \quad (\text{ERJ.18})$$

Their positivity follows from surjectivity of J_N . One actual upper comparator uses the density $\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k - 2 + |u|)^{-B}$ from (ERJ.14). Its even moment of degree d is

$$\int u^d \nu_k(u) du = \frac{2c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} d!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-(k-2)t}}{(\alpha+t)^{d+1}} dt; \quad \int u^d \nu_k(u) du = 0 \quad (d \text{ odd}). \quad (\text{ERJ.19})$$

Insert $(k - 2 + u)^{-B} = (B - 1)!^{-1} \int_0^\infty t^{B-1} e^{-(k-2+u)t} dt$ on the positive half-line; Tonelli and $\int_0^\infty u^d e^{-yu} du = d! y^{-d-1}$ prove the formula. Thus $H_N^- \preceq H_N$ is an explicit strictly positive source. It supplies U_j using comparison moments through $2j$, at most $4q$ at the second endpoint; actual low-degree source data for i stop at $2i \leq 2q$. These cutoffs keep the actual and comparison integrals attached to their stated maps.

Given $0 < L_i \preceq K_i \preceq U_j$, set $\tau = \text{Tr}(L_i^{-1} U_j)$. The Hermitian matrix $L_i^{-1/2} U_j L_i^{-1/2}$ has all q eigenvalues at least one. Each is at most $\tau - q + 1$ because the remaining $q - 1$ contribute at least $q - 1$. It follows that

$$U_j \preceq (\tau - q + 1) L_i, \quad g := \frac{1}{\tau - q + 1} > 0, \quad K_i \succeq g K_j. \quad (\text{ERJ.20})$$

This coefficient is optimal from the trace and the lower bound one: $\text{diag}(\tau - q + 1, 1, \dots, 1)$ attains it. An upper rational $\tau_+ \geq \tau$ gives the certified rational gap $(\tau_+ - q + 1)^{-1}$.

Now take rational Hermitian brackets for both K_i, K_j as in (ERJ.18), and define the exact rational quantities

$$H_* = I - L_i U_j^{-1}, \quad t_m^* = \text{Tr } H_*^m, \quad d = \text{Tr}(U_i L_j^{-1}) - \text{Tr}(L_i U_j^{-1}), \quad r = 1 - g. \quad (\text{ERJ.21})$$

Here g is certified for the reference pair (L_i, U_j) itself. For increasing generalized eigenvalues the exact minimum-maximum formula is

$$\lambda_a(F, G) = \min_{\dim W=a} \max_{0 \neq x \in W} \frac{x^* F x}{x^* G x}. \quad (\text{ERJ.22})$$

For $G = I$, the span of the first a eigenvectors attains λ_a ; every a -dimensional W intersects the span of eigenvectors $a, \dots, q$, whose dimension is $q - a + 1$, and hence its maximum is at least λ_a . The invertible substitution $y = G^{1/2} x$ proves the general case with subspace dimensions preserved.

The pointwise Rayleigh quotients for (L_i, U_j) , (K_i, K_j) , and (U_i, L_j) are ordered. Their extrema are therefore ordered. If $\alpha_a = \lambda_a(L_i, U_j)$ and $\beta_a = \lambda_a(K_i, K_j)$, this proves

$$g \leq \alpha_a \leq \beta_a \leq 1, \quad 0 \leq \sum_a (\beta_a - \alpha_a) \leq d. \quad (\text{ERJ.23})$$

The sum identity uses similarity of FG^{-1} to its positive Hermitian representative. The kernels F, G are forms on the dual quotient coefficient space: $G^{-1}F$ there and FG^{-1} on the quotient are similar through the invertible map G . These β_a are the actual g_a of the return. For $r > 0$, integration of the derivative of x^m on $[0, r]$ gives the complete moment enclosure

$$a_m := \max(0, t_m^* - mr^{m-1}d) \leq s_m := \text{Tr}((H_{i,j}^{\text{loss}})^m) \leq t_m^* \leq qr^m. \quad (\text{ERJ.24})$$

Indeed apply $0 \leq u^m - v^m \leq mr^{m-1}(u - v)$ to $u = 1 - \alpha_a$, $v = 1 - \beta_a$ and sum (ERJ.23). If $r = 0$, all $\alpha_a = \beta_a = 1$ and both loss operators vanish; the exact certificate is zero.

Integrating the geometric series on $[0, x]$ proves $-\log(1 - x) = \sum_{m \geq 1} x^m/m$ for $0 \leq x < 1$. All terms are nonnegative. Its tail is bounded by $x^{p+1}/((p+1)(1-x))$. Thus for every integer $p \geq 1$,

$$\sum_{m=1}^p \frac{a_m}{m} \leq \mathcal{L}_{i,j} \leq \sum_{m=1}^p \frac{t_m^*}{m} + \frac{t_{p+1}^*}{(p+1)g}. \quad (\text{ERJ.25})$$

The width of the two stated endpoints is at most

$$\frac{d(1 - r^p)}{g} + \frac{qr^{p+1}}{(p+1)g}. \quad (\text{ERJ.26})$$

The first term sums $r^{m-1}d$ using (ERJ.24); the second uses $t_{p+1}^* \leq qr^{p+1}$. This accounts for the finite matrix error and series tail separately with their original dimension.

The sharper coefficient

$$c_p(r) = \sum_{l \geq 0} \frac{r^l}{p+1+l} = \frac{-\log g - \sum_{m=1}^p r^m/m}{r^{p+1}} \leq \frac{1}{(p+1)g} \quad (\text{ERJ.27})$$

can replace the last coefficient in (ERJ.25): the tail at $x \leq r$ is at most $c_p(r)x^{p+1}$. To certify its logarithm, write exactly $x = 2^e y$ for rational $x > 0$, integer e , and $1 \leq y < 2$. For $t = (y-1)/(y+1)$, integration of $2/(1-t^2)$ gives

$$\log y = 2 \sum_{a=0}^{M-1} \frac{t^{2a+1}}{2a+1} + E_M, \quad 0 \leq E_M \leq \frac{2t^{2M+1}}{(2M+1)(1-t^2)}. \quad (\text{ERJ.28})$$

At $t = 1/3$ the same formula encloses $\log 2$. Multiply that interval by e , reversing endpoints when $e < 0$, and add the $\log y$ interval. If $\log g \in [\ell_-, \ell_+]$ has width ε , replacing $-\log g$ by $-\ell_-$ in (ERJ.27) adds at most $\varepsilon t_{p+1}^*/r^{p+1} \leq q\varepsilon$ to the upper endpoint. The $r = 0$ case was handled before this division.

3.6 Constructing rational brackets for the actual relation

Let an actual Hermitian source matrix M have a rational Hermitian center M_0 and symmetric rational radii $\varepsilon_{ab} \geq 0$ with $|(M - M_0)_{ab}| \leq \varepsilon_{ab}$. For fixed positive rational w_a set $D_{aa} = \sum_b \varepsilon_{ab} w_b/w_a$. Squaring $\sqrt{t}x - y/\sqrt{t}$ proves $2xy \leq tx^2 + t^{-1}y^2$. Therefore

$$\begin{aligned} |z^*(M - M_0)z| &\leq \sum_{a,b} \varepsilon_{ab} |z_a| |z_b| \\ &\leq \frac{1}{2} \sum_{a,b} \varepsilon_{ab} \left(\frac{w_b}{w_a} |z_a|^2 + \frac{w_a}{w_b} |z_b|^2 \right) = z^* D z. \end{aligned} \quad (\text{ERJ.29})$$

Hence $M_0 - D \preceq M \preceq M_0 + D$. Exact Hermitian elimination tests positivity: a first pivot $d_0 > 0$ gives $d_0|x + d_0^{-1}b^*y|^2 + y^*(C - bb^*/d_0)y$. Induction proves positivity precisely when all successive pivots and their Schur complements are positive.

For the actual, possibly nonrational coefficients of χ , use the exact ordered coefficient matrix

$$\mathcal{A}_N = [1, S, \dots, S^{q-1}, \chi, S\chi, \dots, S^{N-q}\chi]. \quad (\text{ERJ.30})$$

It is triangular with diagonal one. At $N = q - 1$ its relation block is empty. Monic division gives $J_N \mathcal{A}_N = [I_q, 0] =: J_0$. Consequently, in the original remainder coordinates,

$$M_N = \mathcal{A}_N^* H_N \mathcal{A}_N, \quad K_N = J_0 M_N^{-1} J_0^*. \quad (\text{ERJ.31})$$

Intervals for the original source entries and original coefficients of χ enter the literal finite products for M_N ; interval addition, multiplication, and conjugation contain those products. The sum of the real and imaginary radii bounds a complex error. Use conjugate centers and symmetrically enlarged radii, then (ERJ.29) and exact positive pivots. The two inverse compressions of the resulting source brackets yield rational L_N, U_N by (ERJ.18). No center polynomial replaces the actual quotient map in (ERJ.31).

For a fixed actual pair and fixed w_a , refining source and coefficient intervals makes the radii and D tend to zero. Positivity of the actual M_N ensures eventual positive lower pivots. The cofactor formula proves continuity of inversion where the determinant is nonzero; thus $L_N, U_N \rightarrow K_N, d \rightarrow 0$, and the reference gap in (ERJ.20) tends to a positive number. For any requested positive width, first make d/g sufficiently small and then choose p so the geometric tail in (ERJ.26) is smaller than the remaining width. The rational logarithm error can be decreased independently. This proves convergence of the finite algebraic certificate for each fixed actual pair. The quantities controlling that convergence remain explicit when the degree is allowed to grow.

3.7 The four-volume comparison with its literal finite constants

The arithmetic pair losses add to $\mathcal{B}_{h,k} = \log(V_{q-1}V_q/(V_{2q-1}V_{2q}))$. Equations (ERJ.15)–(ERJ.17) prove

$$\mathcal{B}_{h,k} \leq q \log C_{2q-1} + q \log C_{2q}, \quad \frac{C_{2q}}{C_{2q-1}} = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1}-1}{4^{2q}-1} \right)^2 \leq \frac{147}{5} < 32. \quad (\text{ERJ.32})$$

Indeed the first ratio is at most $5/3$, and the ratio inside the square is $4 + 3/(4^{2q} - 1) \leq 21/5$.

The signed metric control now bounds these same two return losses before the scalar envelope is taken. Use $\mathcal{B}_{h,k}^\Gamma$ and $\mathcal{J}_{h,k}$ and the recursively refined endpoints $\mathcal{B}_{h,k}^{\text{lo}}, \mathcal{B}_{h,k}^{\text{hi}}$, as defined and proved in (FV.M1)–(FV.M18), Section 1.1.6. Its common monomial source $H_N(1)$ is exactly the original H_N in (ERJ.3): both have entries $\int (c + iu)^a (c + iu)^b m_k(u) du$. Its quotient J_N is monic remainder by this same χ . If T_N is the monic triangular coefficient matrix of $p_0, \dots, p_N$, then $O_N = T_N^* H_N T_N$, $B_N^{\text{ERJ}} = J_N T_N$, and direct substitution gives $B_N^{\text{ERJ}} O_N^{-1} (B_N^{\text{ERJ}})^* = J_N H_N^{-1} J_N^*$. Hence no change of quotient coordinates occurs in the comparison. In particular the literal refinement of (ERJ.32) is

$$\mathcal{B}_{h,k}^{\text{lo}} \leq \mathcal{B}_{h,k} \leq \min\{q \log C_{2q-1} + q \log C_{2q}, \mathcal{B}_{h,k}^{\text{hi}}\}. \quad (\text{ERJ.32a})$$

The lower bound combines nonnegativity of the two original return losses with (FV.M18); the upper bound combines that equation with (ERJ.32). Each rational trace certificate in (ERJ.25) is retained as well: the two upper endpoints can be summed and included as a further entry in the minimum. The source and relation errors in (ERJ.26)–(ERJ.31) continue to be computed for those original matrices.

This also controls the earlier inverse-power observations, not merely their determinant. Let $g_{a,j}$, $a = 0, 1, 1 \leq j \leq q$, be all eigenvalues of the two original returns, and let

$$U_{h,k}^{\text{joint}} := \min\{q \log C_{2q-1} + q \log C_{2q}, \mathcal{B}_{h,k}^{\text{hi}}\}.$$

They lie in $(0, 1]$, and (ERJ.5) gives the exact identity $\sum_{a,j} -\log g_{a,j} = \mathcal{B}_{h,k}$. For every real $s > 0$, put $z_{a,j} = g_{a,j}^{-s} - 1 \geq 0$. The expansion of $\prod(1 + z_{a,j})$ contains $1 + \sum z_{a,j}$ and only further nonnegative terms. Consequently

$$\begin{aligned} \mathrm{Tr} U_{q-1,2q-1}^{-s} + \mathrm{Tr} U_{q,2q}^{-s} &\leq 2q - 1 + e^{s\mathcal{B}_{h,k}} \leq 2q - 1 + e^{sU_{h,k}^{\mathrm{joint}}}, \\ g_{a,j} &\geq e^{-\mathcal{B}_{h,k}} \geq e^{-U_{h,k}^{\mathrm{joint}}}. \end{aligned} \quad (\text{ERJ.32b})$$

The individual lower bound follows because each nonnegative $-\log g_{a,j}$ is at most their full sum. All powers are of the typed return endomorphisms in (ERJ.4), so (ERJ.11a) identifies them with the inverse singular-value observations of the original restricted lifts. Their action defect and theta primitive remain (ERJ.6)–(ERJ.8). This finite bound is compatible with the central lower estimates below; it does not assign an order-uniform bound to $U_{h,k}^{\mathrm{joint}}$.

Set $X_0 = \max(1, M_h(b), M_h(-b))$, $K_0 = \max(1, \vartheta_h^{-1})$ and $C_{\mathrm{env}} = \max(1, 2/(3c_h D))$. Since $bT \geq 2q$, the factorial ratio in U_k is at most one; also $M_k = 2 \int \cosh(bu) m_k \geq 2\mu_h^k$. Thus $U_k(T) \leq 3X_0^k$. The stated coefficient constant obeys $A_{2q-1}^2 \leq (2q/9)256^q$. On $0 \leq r_0 \leq 1/2$, the derivative of $\log((1+r_0)/(1-r_0))$ is at most $8/3$, so $\log B_q(r_0)^2 \leq 16R_0 k/(3D)$. Substitution into C_{2q-1} , using $\vartheta_h^{-(k-3)} \leq K_0^k$ and $k-2+Dq \leq 1+(D+1)q$, proves

$$\begin{aligned} q \log C_{2q-1} &\leq \mathcal{U}_k, \quad \mathcal{B}_{h,k} \leq 2\mathcal{U}_k + q \log 32, \\ \mathcal{U}_k &= (\alpha D + \log 256)q^2 + \left(\log(X_0 K_0) + \alpha + \frac{16R_0}{3D} \right) kq + Bq \log(1 + (D+1)q) + q \log C_{\mathrm{env}}. \end{aligned} \quad (\text{ERJ.33})$$

All constants are independent of k for the fixed packet and b . As $q = [1 + k(m-1)](k+1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. The displayed scalar majorant divided by q^2 tends to $2(\alpha D + \log 256)$; divided by $q \log k$ it grows. The finite matrix and trace bounds (ERJ.18)–(ERJ.28) observe the same original four volumes and retain the directional data before this scalar estimate.

3.8 Central rank, both endpoint returns, and all inverse powers

Keep the original phase ϕ_N and relative Hermitian radius ϵ_N . For a reflection-stable packet the exact monic recurrence is $Sp_n = p_{n+1} + (c + ia_n)p_n - (\omega_n/\omega_{n-1})p_{n-1}$, $a_n \in \mathbb{R}$. It follows by transporting to u with $p_n(c + iu) = i^n Q_n(u)$ and pairing self-adjoint multiplication by u with every lower monic orthogonal polynomial. Substitution into (ERJ.3) cancels the internal adjacent terms and gives

$$V_N = (b_{N+1}b_N^* + b_N b_{N+1}^*)/\omega_N, \quad b_N^* G_N b_{N+1} = i\omega_N \phi_N. \quad (\text{ERJ.34})$$

The pairing in the second formula is pure imaginary: in the real u coordinates both the quotient Gram and the remainder columns are real, and the source factors contribute $\bar{i}^N i^{N+1} = i$. This defines the retained real phase; it equals the original phase in the monic norm derivative formula. The relative Hermitian matrix has rank at most two and trace zero. Its square has trace $2[(b_N^* G_N b_N)(b_{N+1}^* G_N b_{N+1}) - |b_N^* G_N b_{N+1}|^2]/\omega_N^2$, giving eigenvalues $\epsilon_N, -\epsilon_N$ and the remaining zeros. The rank-one determinant update in (ERJ.3) gives, for $N \geq q$, $b_N^* G_N b_N/\omega_N = 1 - \delta_N$ and $b_{N+1}^* G_N b_{N+1}/\omega_{N+1} = \delta_{N+1}^{-1} - 1$, where $\delta_N = V_N/V_{N-1}$. Therefore

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N)(\delta_{N+1}^{-1} - 1). \quad (\text{ERJ.35})$$

The determinant update follows by eliminating $\begin{pmatrix} I & X \\ -Y & I \end{pmatrix}$ in its two block orders to prove $\det(I + XY) = \det(I + YX)$. Multiplying (ERJ.35) for $N = q, \dots, 2q-2$ proves

$$\prod_{N=q}^{2q-2} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_{2q-1} V_q}{\omega_q V_{2q-1}} (1 - d_0)(1 - d_{q-1}) \prod_{a=1}^{q-2} (1 - d_a)^2, \quad d_a = \delta_{q+a}. \quad (\text{ERJ.36})$$

Each internal contraction occurs twice and both endpoint contractions occur once.

For (ERJ.13) the complete positive-real spectral compression gives

$$L = 2\delta\ell(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad L/q \geq \delta k/2, \quad 0 < L \leq \epsilon_N. \quad (\text{ERJ.37})$$

Here twice the sum of positive real parts of the eigenvalues of $A - cI$ equals L : sum the positive integers $2a - k$, giving $\lfloor (k+1)^2/4 \rfloor$, then include all $k+1$ imaginary centres and their full length ℓ . A unitary upper triangular form in the actual G_N metric, with these eigenvalues first, shows that this is the trace of the Hermitian defect compressed to their invariant subspace. A projection paired with a Hermitian matrix of eigenvalues $\epsilon_N, -\epsilon_N, 0, \dots, 0$ has trace at most ϵ_N , proving the stated bound. The integer inequality $\lfloor (k+1)^2/4 \rfloor \geq k(k+1)/4$ follows in each parity.

The complete balanced monic-norm source proves

$$\begin{aligned} (\omega_{n+r}/\omega_n)^{1/(2r)} &\leq C_h^{\text{bal}} n \quad (n \geq k \geq 3, n/2 \leq r \leq n), \\ C_h^{\text{bal}} &= 256e^\pi \max(1, 3/c_h) X_0 K_0 \max(1, b^{-3}) 6^{B/3}. \end{aligned} \quad (\text{ERJ.38})$$

This is its literal sufficient constant from the original envelope and monic Legendre minimum, with its full analytic proof included in the dependency record. Apply it at $n = q, r = q - 1$ and put $X = L/(C_h^{\text{bal}} q)$. Every d_a in (ERJ.36) is strictly below one, since otherwise its right side is zero while every left factor is at least $L^2 > 0$. Define the entire nonnegative penalty

$$\begin{aligned} P_{\text{mid}} &= -\log(1 - d_0) - \log(1 - d_{q-1}) - 2 \sum_{a=1}^{q-2} \log(1 - d_a) \\ &\quad + \sum_{a=0}^{q-2} \log(1 + \phi_{q+a}^2/L^2). \end{aligned} \quad (\text{ERJ.39})$$

Taking logarithms in (ERJ.36) proves

$$C_{\text{mid}} := \log(V_q/V_{2q-1}) \geq 2(q-1) \log X + P_{\text{mid}}. \quad (\text{ERJ.40})$$

Indeed every factor $\epsilon_N^2 + \phi_N^2$ is at least $L^2(1 + \phi_N^2/L^2)$, and (ERJ.38) bounds the norm ratio.

The central update is the full matrix

$$K_{2q-1} - K_q = \sum_{a=q+1}^{2q-1} b_a b_a^* / \omega_a, \quad \text{rank}(K_{2q-1} - K_q) \leq q - 1. \quad (\text{ERJ.41})$$

Its positive Hermitian representative shows that the central return has at least one exact unit eigenvalue. List one such value separately and the other $q - 1$ as $\gamma_1, \dots, \gamma_{q-1}$, keeping any additional unit values. Their product is $e^{-C_{\text{mid}}}$. For every real $s > 0$, convexity of e^{sx} gives

$$\begin{aligned} \text{Tr } T_{q,2q-1}^{-s} &= 1 + \sum_{a=1}^{q-1} \gamma_a^{-s} \\ &\geq 1 + (q-1)e^{sC_{\text{mid}}/(q-1)} \\ &\geq \max \left\{ q, 1 + (q-1)X^{2s}e^{sP_{\text{mid}}/(q-1)} \right\}. \end{aligned} \quad (\text{ERJ.42})$$

For completeness $e^y \geq 1 + y$ follows by differentiating $e^y - 1 - y$, whose minimum is zero at $y = 0$. Apply it to $y_a = -s \log \gamma_a - \bar{y}$, where $\bar{y} = -(s/(q-1)) \sum_a \log \gamma_a$: $\sum_a e^{-s \log \gamma_a} = e^{\bar{y}} \sum_a e^{y_a} \geq (q-1)e^{\bar{y}}$. The q bound also follows termwise from $\gamma_a \leq 1$. The actual rank $\rho = \text{rank}(K_{2q-1} - K_q)$ preserves exactly $q - \rho$ unit eigenvalues, because $I - T = (K_{2q-1} - K_q)G_{2q-1}$ has rank ρ . Write

$A_+ = \max\{0, 2(q-1) \log X + P_{\text{mid}}\}$. If $\rho > 0$, applying the same exponential argument to just the ρ eigenvalues below one proves the stronger exact-rank consequence

$$\text{Tr } T_{q,2q-1}^{-s} \geq q - \rho + \rho e^{sC_{\text{mid}}/\rho} \geq q - \rho + \rho e^{sA_+/\rho}. \quad (\text{ERJ.42a})$$

For fixed $a \geq 0$, $r(e^{a/r} - 1)$ decreases as $r > 0$ increases: its derivative is $e^{a/r}(1 - a/r) - 1 \leq 0$, since the derivative of $e^t(1 - t)$ is $-te^t \leq 0$ for $t \geq 0$. Thus $\rho \leq q - 1$ proves that (ERJ.42a) improves (ERJ.42). At $\rho = 0$ the central return is the identity, its cost is zero, and the trace is exactly q . There is no division by ρ in that case. By (ERJ.22), the ordered eigenvalues of each pair (K_{q-1}, K_{2q-1}) and (K_q, K_{2q}) are at most those of (K_q, K_{2q-1}) : their Rayleigh quotients decrease respectively in the numerator and by increase of the denominator. Since x^{-s} is decreasing, *each* original endpoint return obeys the final bound in (ERJ.42), and the stronger rank-refined bound (ERJ.42a), by the same ordered comparison. Its minimum eigenvalue also obeys

$$g_{\min}^{(a)} \leq \min \left\{ 1, X^{-2} e^{-P_{\text{mid}}/(q-1)} \right\} \leq \min \left\{ 1, \frac{4(C_h^{\text{bal}})^2}{\delta^2 k^2} \right\}, \quad a = 0, 1. \quad (\text{ERJ.43})$$

The first inequality follows from $\gamma_{\min}^{q-1} \leq \prod \gamma_a$ and (ERJ.40), followed by the same ordered comparison. Equations (ERJ.9)–(ERJ.12) and (ERJ.7)–(ERJ.8) give its exact source, action, and theta interpretation. The identity

$$\mathcal{B}_{h,k} = 2C_{\text{mid}} + \log(V_{q-1}/V_q) + \log(V_{2q-1}/V_{2q}) \quad (\text{ERJ.44})$$

has nonnegative endpoint terms. Combining (ERJ.37)–(ERJ.40) proves

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4,$$

because its lower bound is $4(q-1)[\log k + \log(\delta/(2C_h^{\text{bal}}))]$ and $(q-1)/q \rightarrow 1$. The consecutive-first-window continuation separately proves the length- q products at the two actual endpoints, including the single terminal factor at initial degree $q-1$, and gives its stronger per-window inverse-power estimate. Its original constant and phase formula stay in that complete companion proof; (ERJ.42) retains the exact central unit eigenvalue and full central penalty.

3.9 Exact auxiliary calculation and retained proof dependencies

The accompanying complete fixture proof uses $S = 1 + iu$, $\chi = S^2 - 2S - 1$ and $r_\lambda(u) = |\Gamma(\lambda + iu/2)|^2/(2\pi)$. Its source is $m = r_4/70$ of mass $9/16$; its smaller source is $\nu = (54/35)(r_2/3)$ of mass $27/70$. The full identity

$$m(u) = \left(\frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120} \right) \frac{r_2(u)}{3} \quad (\text{ERJ.45})$$

follows from two applications of $\Gamma(z+1) = z\Gamma(z)$, and proves their source contraction. The complete fixture appendix derives all moments, original S Grams, remainder matrices, extrema, and rational logarithm endpoints. Its remainder norm squared is $763/108$, attained by the original polynomial $-245/36 + (791/108)S - (7/9)S^2 + (7/27)S^3$ in both degree three and degree four. The resulting source gap is $108/763$, while the two actual minimum return eigenvalues equal $135/331$. The exact four-volume ratio is $3073295611/216513000$. These quantities are proved in the full fixture text retained with this article, rather than inferred from a decimal computation.

The existing fixture executed 303 distinct mathematical checks in each unchanged normal and optimized run. Seven literal formula changes in each mode were rejected after the complete calculation, giving 16 actual jobs in total. The preserved source and full execution JSON have hashes recorded in this article's manifest. This completion did not rerun that unchanged suite. Its exact source/Gamma fixture is connected to the arithmetic construction by (ERJ.2)–(ERJ.4), (ERJ.17)–(ERJ.20); the density (ERJ.45) and its masses remain its declared auxiliary inputs.

For the empty packet $h = 1$, $E = 0$, the quotient determinants are empty determinants one and every quotient trace and four-volume loss is zero. The original analytic source and mass still exist. The positive inverses and $q - 1$ spectral averages above are used only in their stated domains; the actual quartet has $q \geq (k + 1)^2$ and $k \geq 3$. At $i = j$, return is the identity, its relation correction and projection difference vanish, and the exact loss is zero.

The complete source dependencies are the original *Canonical restriction and the arithmetic endpoint volume*; *An arithmetic lower-envelope comparison for central quotient volumes* (AU.1–49); *Exact window products, sharp endpoint optimization, and original relation costs* (EP.1–54), with its complete balanced-window proof dependency; the endpoint-window proof (EW.1–59); the complete 303-check fixture proof; and the *Consecutive first-window join* (CJ.1–22). The accompanying manifest pins their actual files and the new independent mathematical review. The global cumulative reader carries these full proofs and the new article in their stated order.

4 Recursive propagation of the original signed metric bounds

This proof combines the earlier full-relative-spectrum estimate AW16, the exact trace norm and actual-overlap estimates of SP, the supplied centered Hilbert–Schmidt estimate, and the all-degree angle estimate EP17–EP25 on their common original source. Each of those complete proofs is retained in this edition. The construction below carries their pointwise minimum through both the original volume and the exact nonlinear endpoint coordinate, without changing either endpoint form.

4.1 The original source and all four quotient volumes

Fix the original packet h , tensor order $k \geq 1$, monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$, and

$$H = \mathcal{P}_{2q}, \quad S = k/2 + iy, \quad v(y) = (1, S, \dots, S^{2q}). \quad (\text{RMT1})$$

The coordinates are the original ascending monomial coordinates. The original reference and arithmetic measures are

$$d\mu_0 = r_{1/4}^{*k}(y) dy, \quad d\mu_1 = w_h^{*k}(y) dy, \quad w_h(y) = \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi}. \quad (\text{RMT2})$$

All their original mass and Taylor-unit factors remain. In particular the exact reference convolution identity is

$$r_{1/4}^{*k} = \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}, \quad c_a = 2^{1-2a} \Gamma(2a), \quad c_{1/4} = \sqrt{2\pi}. \quad (\text{RMT3})$$

Thus these are the same measures in AW and SP. Every argument below also applies to the actual finite-circle comparison when one of the measures is its specified positive atomic measure on this same line and its retained degree- $2q$ Gram is positive definite. That positivity is already proved by the original sampled-source bounds. This statement does not assign moment structure to an arbitrary positive coefficient matrix.

For $0 \leq x \leq 1$ let

$$M_x = \int v(y)^* v(y) d((1-x)\mu_0 + x\mu_1)(y), \quad \dot{M} = M_1 - M_0, \quad C_x = M_x^{-1} \dot{M}. \quad (\text{RMT4})$$

The two positive endpoint forms make M_x positive definite throughout the closed interval. For every original cutoff N , let $I_N : \mathcal{P}_N \rightarrow H$ be coefficient inclusion and $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ , followed by inclusion. Negative relation degree means the zero space. Retain the orthogonal projections

$$P_N = I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \quad Q_N = B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0. \quad (\text{RMT5})$$

AW's relation matrix has target $\mathcal{P}_N$, so its exact relation to this one is $B_N = I_N B_N^{\text{AW}}$. Equation (RMT5), after that substitution, is precisely AW4 and SP2. The quotient map remains $J_N : \mathcal{P}_N \rightarrow E = \mathbb{C}[S]/(\chi)$, and its canonical positive quotient Gram is $G_N(x)$. Put $V_N(x) = \det G_N(x)$ and retain the four signs:

$$\mathcal{B}(x) = \log \frac{V_{q-1}(x)V_q(x)}{V_{2q-1}(x)V_{2q}(x)}, \quad U_x = Q_{2q-1} + Q_{2q} - Q_q, \quad W_x = P_{2q-1} - P_{q-1} + P_{2q} - P_q. \quad (\text{RMT6})$$

The relation-first determinant-line formula and differentiation of each of these four determinants give, without commuting their operators,

$$\mathcal{B}'(x) = \text{Tr}((U_x - W_x)C_x). \quad (\text{RMT7})$$

This is the same actual integrand as AW6 and SP7.

4.2 The simultaneous bounds on that integrand

Let $b_1 \leq \dots \leq b_{2q+1}$ be every eigenvalue, with multiplicity, of $D = M_0^{-1}M_1$ in the original M_0 metric. They are positive. Retain

$$c_j(x) = \frac{b_j - 1}{1 - x + xb_j}, \quad d_x = c_{2q+1}(x) - c_1(x), \quad \kappa = \frac{b_{2q+1}}{b_1}. \quad (\text{RMT8})$$

These are the ordered eigenvalues of C_x . Indeed $C_x = ((1-x)I + xD)^{-1}(D - I)$, and differentiation with respect to b_j gives the positive quantity $(1 - x + xb_j)^{-2}$. In particular $\int_0^1 d_x dx = \log \kappa$.

Set $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. The two ranks are $q + 1$ in dimension $2q + 1$, so $s_x \geq 1$. All traces and trace norms below are in the same original M_x metric. Define the two calculated functions

$$E_q(x) = c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \quad (\text{RMT9})$$

$$O_q(x) = \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}.$$

The sum is empty when $q = 1$. The complete nested-flag calculation gives both U_x, W_x the spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. Decomposing each as the sum of its spectral projections of ranks $q + 1$ and $q - 1$, and applying the ordered projection-trace inequalities, gives $|\mathcal{B}'| \leq E_q$, exactly as in AW15-AW16. For clarity the subtraction of the two bounds cancels the common c_{q+1} only after both sums have been written; it leaves every term displayed in E_q .

Both positive operators dominate the original orthogonal projection onto their common range. Subtracting that same projection from both leaves positive operators with traces $2q - s_x$, whence $\|U_x - W_x\|_1 \leq 4q - 2s_x$. Also their computed spectra give

$$\text{Tr}((U_x - W_x)^2) = 8q - 4 - 2\text{Tr}(U_x W_x). \quad (\text{RMT10})$$

Cauchy-Schwarz on all $2q + 1$ eigenvalues of $U_x - W_x$ gives the second trace-norm bound in (RMT9). Since $\text{Tr}(U_x - W_x) = 0$, subtract $(c_1 + c_{2q+1})I/2$ from C_x in (RMT7). In an M_x -orthonormal eigenbasis of $U_x - W_x$, the absolute value of each diagonal entry of this centered C_x is at most $d_x/2$. Multiplying by the corresponding absolute eigenvalues and summing gives the retained exact trace-norm bound

$$|\mathcal{B}'| \leq T_q(x) := \frac{d_x}{2} \|U_x - W_x\|_1 \leq O_q(x). \quad (\text{RMT10a})$$

The second inequality follows from the two trace-norm estimates just proved. Thus neither of those majorants replaces the original trace norm.

The user's additional supplied projector calculation gives another bound on this same integrand. Define

$$\Sigma_x = \text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{2q+1}, \quad S_q(x) = \sqrt{\Sigma_x \text{Tr}((U_x - W_x)^2)}. \quad (\text{RMT10b})$$

The nonnegative number Σ_x is the squared Hilbert–Schmidt norm of $C_x - \text{Tr}(C_x)I/(2q+1)$ in the original M_x Hilbert space. The trace of $U_x - W_x$ is zero, so Hilbert–Schmidt Cauchy–Schwarz gives $|\mathcal{B}'| \leq S_q$. Equivalently one may apply the isometry $M_x^{1/2} : (H, M_x) \rightarrow (H, I)$ simultaneously to both operators; conjugation preserves their traces and products and proves the same inequality. This is a calculation with the original metric, not a change of either endpoint source form.

The original crossed quotient pairs are $(q-1, 2q)$ and $(q, 2q-1)$. Retain their full canonical minimum-section angle lists $(\theta_1, \dots, \theta_q)$ and $(\varphi_1, \dots, \varphi_q)$. In the second list, ordered with a smallest angle first, the original ambient dimension forces $\varphi_1 = 0$. The count is proved in EP17–EP25. For the following scalar sums put $a = 2q-1$ and define the explicitly indexed list

$$\lambda_i = -\log \cos^2 \theta_i \quad (1 \leq i \leq q), \quad \lambda_{q+j-1} = -\log \cos^2 \varphi_j \quad (2 \leq j \leq q).$$

Then $\mathcal{B} = -\log \cos^2 \varphi_1 + \sum_{\nu=1}^a \lambda_\nu = 0 + \sum_{\nu=1}^a \lambda_\nu$. Every additional zero remains in the λ -list, while the designated geometric entry φ_1 is still displayed in the full second list. The exact projection-difference spectra give

$$|\mathcal{B}'(x)| \leq d_x \left(\sin \varphi_1 + \sum_{\nu=1}^a \sqrt{1 - e^{-\lambda_\nu(x)}} \right) \leq a \sqrt{1 - e^{-\mathcal{B}(x)/a}} d_x. \quad (\text{RMT11})$$

The second inequality is concavity of $z \mapsto \sqrt{1 - e^{-z}}$ on $[0, \infty)$, applied to the full a -term list, including its zeros, together with the displayed $\sin \varphi_1 = 0$. Both q -entry lists remain the geometric data; the known zero contribution is kept explicitly in these scalar identities.

The needed strict positivity of $\mathcal{B}$ follows on the actual measures rather than being an additional assumption. If $\mathcal{B} = 0$, both nested quotient pairs have equal minimum spaces. To see every implication, their canonical minimum sections satisfy $G_i - G_j = (R_i - R_j)^* M_x (R_i - R_j) \succeq 0$. Thus each determinant ratio in the crossed pairs is at least one. A zero sum of their nonnegative logarithms makes both ratios one. All eigenvalues of $G_j^{-1/2} G_i G_j^{-1/2}$ are at least one, so determinant one makes every eigenvalue one and $G_i = G_j$. The displayed squared norm then gives $R_i = R_j$ as maps to the same original source H . In the first pair R_{q-1} is the literal degree-less-than- q remainder section. Its equality with R_{2q} puts the constant polynomial 1 in the latter minimum space, which is orthogonal to $\ker J_{2q} = \chi \mathcal{P}_q$. For $\chi(S) = \sum a_j S^j$, put $\chi^{\#k}(S) = \sum \bar{a}_j (k - S)^j$. This belongs to $\mathcal{P}_q$ and equals $\overline{\chi(S)}$ on $S = k/2 + iy$. The asserted orthogonality would therefore give

$$0 = \int \chi(S) \chi^{\#k}(S) d\mu_x = \int |\chi(S)|^2 d\mu_x, \quad (\text{RMT12})$$

contradicting the positive Gram of the nonzero polynomial χ . This proof also covers the specified positive atomic comparison. Thus $\mathcal{B}(x) > 0$ on the entire closed interval.

4.3 Carrying the refined bound through both endpoint coordinates

Define the positive function

$$z_x = a \sqrt{1 - e^{-\mathcal{B}(x)/a}} > 0.$$

With every preceding control retained, set

$$\begin{aligned} A_q(x) &= z_x d_x, \\ J_q &= \int_0^1 \min\{E_q(x), T_q(x), O_q(x), S_q(x), A_q(x)\} dx, \\ I_q &= \int_0^1 \min\left\{d_x, \frac{E_q(x)}{z_x}, \frac{T_q(x)}{z_x}, \frac{O_q(x)}{z_x}, \frac{S_q(x)}{z_x}\right\} dx. \end{aligned} \quad (\text{RMT13})$$

These are actual quantities of the original finite source path. The projections are continuous matrix functions of x . Explicitly $s_x = 2(q+1) - \text{rank}[U_x W_x]$. A rank superlevel set is a finite union of sets where a specified minor is nonzero, hence is Borel. This proves measurability of s_x and of every displayed integrand. Equation (RMT12) and compactness give a positive lower bound for $\mathcal{B}$ on this fixed path. Hence all denominators in I_q are nonzero, and the integrands are finite and bounded. This is a fixed-source statement and asserts no bound uniform in the tensor order.

All five preceding estimates bound the same $|\mathcal{B}'|$. Pointwise comparison followed by the triangle inequality for the integral therefore proves

$$|\mathcal{B}(1) - \mathcal{B}(0)| \leq J_q, \quad J_q \leq \min \left\{ \mathcal{L}_q(D), \int_0^1 T_q(x) dx, \int_0^1 O_q(x) dx, \int_0^1 S_q(x) dx, \int_0^1 A_q(x) dx \right\}. \quad (\text{RMT14})$$

where the earlier AW13 quantity is retained in full:

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log \kappa. \quad (\text{RMT15})$$

Indeed integration of each c_j gives $\log b_j$, which proves the first entry of the minimum in (RMT14).

For $B > 0$ let

$$\mathcal{H}_a(B) = 2 \operatorname{arcosh}(e^{B/(2a)}), \quad \mathcal{H}'_a(B) = \frac{1}{a\sqrt{1 - e^{-B/a}}}, \quad \mathcal{H}_a^{-1}(z) = 2a \log \cosh(z/2) \quad (z > 0). \quad (\text{RMT16})$$

The derivative follows by differentiating the positive real inverse branch of cosh; substitution proves both inverse composites. Applying this exact derivative to each of the five bounds on $\mathcal{B}'$ gives

$$|\mathcal{H}_a(\mathcal{B}(1)) - \mathcal{H}_a(\mathcal{B}(0))| \leq I_q \leq \int_0^1 d_x dx = \log \kappa. \quad (\text{RMT17})$$

Thus the exact trace norm, supplied variance estimate, overlap and full-relative-spectrum information propagate into the nonlinear bound, not only into its linear consequence.

Writing $H_0 = \mathcal{H}_a(\mathcal{B}(0))$, the inverse in (RMT16), extended continuously by value zero at zero, and the additive estimate in (RMT14) give the simultaneous interval

$$\begin{aligned} \mathcal{B}(1) &\geq \max \left\{ 0, \mathcal{B}(0) - J_q, 2a \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right) \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ \mathcal{B}(0) + J_q, 2a \log \cosh \left(\frac{H_0 + I_q}{2} \right) \right\}. \end{aligned} \quad (\text{RMT18})$$

The actual $\mathcal{B}(1)$ is strictly positive although the lower bound can equal zero. If $M_1 = cM_0$ with $c > 0$, all the c_j coincide, so $d_x, E_q, T_q, O_q, S_q, A_q, I_q, J_q$ vanish. Each original quotient determinant is multiplied by the same power q of $1 - x + xc$; the four signs then give $\mathcal{B}(1) = \mathcal{B}(0)$, as (RMT18) also states. This includes $q = 1$, for which the ordered sums above are empty and $a = 1$.

Every projection and metric in this argument uses the original J_N, B_N , coefficient basis and source measure. The arithmetic class map $\sigma_h^{\otimes k} \eta_{h,k} J_N$, its complete Taylor unit, and the original theta primitives of differences of minimum sections are unchanged. The proof refines numerical bounds on those exact maps; it introduces no identification between their source kernels or between distinct support labels. The earlier complete AW, SP and EP proofs, together with the explicit correspondences (RMT3)–(RMT7), provide the full input proofs used in this propagation.

4.4 Carrying the constructed signed residual into the endpoint

The complete supplied projector correspondence and its explicit finite approximation are proved in Section 5, (ISM14)–(ISM37). Their notation has the exact identities

$$X_x = C_x, \quad \mathcal{A}_x = U_x - W_x, \quad \Pi_N = P_N - Q_N.$$

For completeness retain the actual finite construction here. For each integer $L \geq 0$, put

$$\begin{aligned} B_x^\circ &= (1-x)I + xD, & \alpha_x &= \frac{2(1-x) + x(b_1 + b_{2q+1})}{2}, \\ H_x^\circ &= I - B_x^\circ/\alpha_x, & C_x^{[L]} &= \alpha_x^{-1} \sum_{r=0}^L (H_x^\circ)^r (D - I), \\ R_x^{[L]} &= (H_x^\circ)^{L+1} C_x, & E_x^{[L]} &= R_x^{[L]} - \frac{\text{Tr } R_x^{[L]}}{2q+1} I. \end{aligned} \tag{RMT19}$$

The symbols $B_x^\circ, H_x^\circ, \alpha_x$ label operators and the scalar in this calculation; they leave every original relation matrix B_N , source metric M_x , and coordinate in (RMT1) in place. Multiplication of the finite geometric sum by $I - H_x^\circ$ gives $C_x - C_x^{[L]} = R_x^{[L]}$ exactly. Every factor is a real function of the same M_0 -self-adjoint D , hence is self-adjoint in $M_x = M_0 B_x^\circ$ as well. Since the trace of $U_x - W_x$ vanishes, (RMT7) becomes

$$\mathcal{B}'(x) = \text{Tr}(C_x^{[L]}(U_x - W_x)) + \text{Tr}(E_x^{[L]}(U_x - W_x)).$$

Apply Hilbert–Schmidt Cauchy–Schwarz in the original M_x metric to the last term and integrate. Define the specified real numbers

$$\begin{aligned} j_L &= \int_0^1 \text{Tr}(C_x^{[L]}(U_x - W_x)) dx, \\ e_L &= \int_0^1 \sqrt{\text{Tr}((E_x^{[L]})^2) \text{Tr}((U_x - W_x)^2)} dx, \\ \vartheta &= \frac{b_{2q+1} - b_1}{b_{2q+1} + b_1}, & K_D &= \sum_{j=1}^{2q+1} \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \end{aligned} \tag{RMT20}$$

These are $J_L, \epsilon_L, \vartheta, K_D$ of (ISM34)–(ISM37) with j_L, e_L used to avoid collision with J_q in (RMT13). The eigenvalues of H_x° have absolute value at most $\vartheta < 1$; those of C_x are the full $c_j(x)$. Thus $\text{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$, because subtracting the trace mean subtracts a nonnegative square from the full sum of squared residual eigenvalues. The two positive operators U_x, W_x dominate their range projections. Those ranges intersect in dimension at least one, so $\text{Tr}(U_x W_x) \geq 1$, by positivity of the trace product and the squared-length formula for the product of the range projections. Equation (RMT10) gives $\text{Tr}((U_x - W_x)^2) \leq 8q - 6$. The signed estimate and its computed residual are therefore

$$\begin{aligned} j_L - e_L &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_L + e_L, \\ 0 \leq e_L &\leq \vartheta^{L+1} \sqrt{K_D(8q-6)} \xrightarrow{L \rightarrow \infty} 0. \end{aligned} \tag{RMT21}$$

Every integral is a continuous finite matrix expression. This proves convergence for the fixed actual two source forms; no uniformity in k is assumed or inferred.

Let $B_0 = \mathcal{B}(0)$ and retain H_0 from (RMT18). Intersecting the proved intervals, for every $L \geq 0$, gives the propagated endpoint bounds

$$\begin{aligned} \mathcal{B}(1) &\geq \max \left\{ 0, B_0 - J_q, 2a \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right), B_0 + j_L - e_L \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ B_0 + J_q, 2a \log \cosh \left(\frac{H_0 + I_q}{2} \right), B_0 + j_L + e_L \right\}. \end{aligned} \tag{RMT22}$$

Indeed each lower entry has separately been proved no larger than the actual endpoint, and each upper entry no smaller. Taking their maximum and minimum therefore preserves both inequalities and yields a nonempty interval containing that endpoint. A numerically integrated center must retain its own integration error before use; (RMT20)–(RMT22) state the exact integrals. The scalar source case has $H_x^\circ = E_x^{[L]} = 0$ and $j_L = e_L = 0$, so it also satisfies (RMT22) exactly.

5 Source enclosures and convergent signed projector control

This chapter incorporates the complete written derivations in the user-supplied continuation following merged PR29. The canonical secant and quotient-variation identities already occur in that PR. The four-projector pairing is the same pairing as AW and SP, with the exact coordinate correspondence proved below. The source sandwich, restriction-loss expression for its Hilbert–Schmidt size, and centered signed enclosure are retained in their full forms. We additionally construct the approximating operator in that enclosure and prove an explicit convergent residual bound. The supplied Lean draft is preserved separately as uncompiled; the mathematical scope here is the written proof.

5.1 The unchanged polynomial and arithmetic source

Fix the actual full packet h , tensor order $k \geq 1$, and its monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Put $E = \mathbb{C}[S]/(\chi)$, $H = \mathcal{P}_{2q}$, and $n = 2q + 1$. All coordinates are the original ascending monomial coordinates. The original analytic source and observation are

$$\begin{aligned} g &= 2\xi, & \mathcal{M}F_h &= g/h, & \mathcal{V}P &= P(D_1 + \cdots + D_k)F_h^{\otimes k}, \\ J^{(k)}\mathcal{V}P &= \eta_{h,k}\pi_\chi P, & \eta_{h,k}[P] &= v_h^{\otimes k} P(A_h^{(k)})1, & v_h &= j_h(g/h), & A_h^{(k)} &= \sum_{i=1}^k A_{h,i}. \end{aligned} \quad (\text{ISM1})$$

Here A_h is multiplication by the original coordinate on $E_h = \mathbb{C}[s]/(h)$. Thus the domain of $\eta_{h,k}$ is E , its codomain is $E_h^{\otimes k}$, and its full unit is retained. The minimal polynomial definition of χ on the cyclic vector 1 proves that $[P] \mapsto P(A_h^{(k)})1$ is injective; multiplication by the invertible $v_h^{\otimes k}$ preserves injectivity.

For $N \in \{q-1, q, 2q-1, 2q\}$, let $I_N : \mathcal{P}_N \hookrightarrow H$ be coefficient inclusion, $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ followed by inclusion, and $C : E \rightarrow H$ the degree-less-than- q remainder lift in the retained quotient basis. The relation space is zero when $N = q-1$. Polynomial division by the monic χ proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times\chi} \mathcal{P}_N \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (\text{ISM2})$$

Indeed division writes every polynomial uniquely as $\chi Q + r$, $\deg r < q$; if its class is zero, $r = 0$. This also proves injectivity of the relation map and every dimension used below.

Let M_0, M_1 be the actual positive definite Hermitian forms on this same H . In the arithmetic comparison these are the moments of $r_{1/4}^{*k}(y) dy$ and $w_h^{*k}(y) dy$ in $S = k/2 + iy$, with $w_h(y) = |(g/h)(1/2 + iy)|^2/(2\pi)$. The finite identities below also hold for any two positive forms on this unchanged presentation. The canonical section, its relation coefficient, and its positive quotient Gram are

$$\begin{aligned} Z_N(M) &= (B_N^* M B_N)^{-1} B_N^* M C, \\ \rho_N(M) &= C - B_N Z_N(M) : E \longrightarrow H, \\ G_N(M) &= \rho_N(M)^* M \rho_N(M), & V_N(M) &= \det G_N(M) > 0. \end{aligned} \quad (\text{ISM3})$$

At $N = q - 1$ the first map is the unique map to the zero space and $B_N Z_N = 0$; no preceding inverse is used. For every other degree, injectivity of B_N and positivity of M make its relation Gram positive definite. Direct substitution gives $B_N^* M \rho_N = 0$ and $\pi_\chi \rho_N = 1_E$. Hence ρ_N is injective, which proves positivity of G_N .

5.2 Both scaled secants, with the original boundary terms

Fix one admitted N , write $\rho_a = \rho_N(M_a)$, and put

$$\Delta = \rho_1 - \rho_0 = B_N(Z_N(M_0) - Z_N(M_1)). \quad (\text{ISM4})$$

For every coefficient map $Z : E \rightarrow \mathcal{P}_{N-q}$, the two cross terms in the Gram expansion vanish by $B_N^* M_a \rho_a = 0$, giving

$$(\rho_a + B_N Z)^* M_a (\rho_a + B_N Z) = G_N(M_a) + (B_N Z)^* M_a (B_N Z). \quad (\text{ISM5})$$

Taking $a = 0$, $B_N Z = \Delta$, and then $a = 1$, $B_N Z = -\Delta$, proves

$$\rho_1^* M_0 \rho_1 = G_0 + \Delta^* M_0 \Delta, \quad \rho_0^* M_1 \rho_0 = G_1 + \Delta^* M_1 \Delta. \quad (\text{ISM6})$$

Here $G_a = G_N(M_a)$. Subtracting the displayed matrices without deleting either boundary term yields, for real a, b ,

$$\begin{aligned} G_1 - aG_0 &= \rho_1^* (M_1 - aM_0) \rho_1 + a\Delta^* M_0 \Delta, \\ bG_0 - G_1 &= \rho_0^* (bM_0 - M_1) \rho_0 + \Delta^* M_1 \Delta. \end{aligned} \quad (\text{ISM7})$$

In particular both boundary terms have the positive signs shown. For $0 < a \leq b$ and $aM_0 \preceq M_1 \preceq bM_0$, congruence of positive matrices and (ISM7) give

$$aG_N(M_0) \preceq G_N(M_1) \preceq bG_N(M_0) \quad \text{at every one of the four degrees.} \quad (\text{ISM8})$$

Such finite endpoints always exist: they may be taken to be the smallest and largest eigenvalues of $M_0^{-1} M_1$, which is positive self-adjoint in the original M_0 metric.

Conjugating (ISM8) by $G_N(M_0)^{-1/2}$ places every eigenvalue of the conjugated positive matrix in $[a, b]$. The determinant is its product and logarithm is the sum of the real logarithms, so

$$q \log a \leq \log \frac{V_N(M_1)}{V_N(M_0)} \leq q \log b. \quad (\text{ISM9})$$

Set

$$\mathcal{B}(M) = \log \frac{V_{q-1}(M) V_q(M)}{V_{2q-1}(M) V_{2q}(M)}, \quad \mathcal{C} = \mathcal{B}(M_1) - \mathcal{B}(M_0). \quad (\text{ISM10})$$

There are exactly two positive and two negative terms in $\mathcal{C}$. Applying the upper endpoint in (ISM9) to the positive terms and the lower endpoint to the negative ones, and then reversing these choices, proves

$$|\mathcal{C}| \leq 2q \log(b/a). \quad (\text{ISM11})$$

Consequently a certified common source enclosure $(1 - \varepsilon)\widehat{M} \preceq M \preceq (1 + \varepsilon)\widehat{M}$, $0 \leq \varepsilon < 1$, gives

$$|\mathcal{B}(M) - \mathcal{B}(\widehat{M})| \leq 2q \log \frac{1 + \varepsilon}{1 - \varepsilon}. \quad (\text{ISM12})$$

These are the supplied source-error estimates in their original forms. At these same use sites the stronger previously proved AW/SP estimate must also be retained. For all ordered eigenvalues $\beta_1 \leq \dots \leq \beta_{2q+1}$ of $D = M_0^{-1} M_1$, define its full existing expression

$$\mathcal{L}_q(D) = \log \frac{\beta_{q+2}}{\beta_q} + 2 \sum_{j=1}^{q-1} \log \frac{\beta_{2q+2-j}}{\beta_j}.$$

The sum is empty at $q = 1$. AW's two nested-flag trace estimates, with the exact operator identification (ISM16)–(ISM19), give

$$|C| \leq \mathcal{L}_q(D) \leq (2q - 1) \log(b/a).$$

For the last inequality every displayed eigenvalue ratio is at most b/a , and its total coefficient is $1 + 2(q - 1) = 2q - 1$. Thus the same source enclosure in (ISM12) also gives the stronger $(2q - 1) \log((1 + \varepsilon)/(1 - \varepsilon))$ bound.

For the actual moment paths of RMT, let J_q be its integral of the pointwise minimum of the full-spectrum, exact trace-norm, overlap, centered Hilbert–Schmidt, and angle bounds. Each bounds the identical derivative (ISM19); the Hilbert–Schmidt member is (ISM26). Taking their pointwise minimum before integrating gives the recursively refined source-error statement

$$\begin{aligned} |C| &\leq \min\{J_q, \mathcal{L}_q(D), (2q - 1) \log(b/a)\}, \\ C &\in [J_L - \epsilon_L, J_L + \epsilon_L] \cap [-J_q, J_q] \quad (L \geq 0). \end{aligned}$$

Here J_L, ϵ_L are the fully constructed signed center and residual in (ISM31)–(ISM37). Their interval therefore retains its original orientation in this earlier source-error calculation. For generic positive coefficient forms, the full-spectrum, trace-norm, overlap, Hilbert–Schmidt and signed bounds still apply, but the moment-specific angle improvement and its strict moment-positivity proof are used only on the original moment paths. The exact primitive and coefficient support are established in (ISM38)–(ISM43). Scaling a source by $c > 0$ leaves Z_N, ρ_N unchanged and multiplies G_N, V_N by c, c^q , respectively, directly by (ISM3). Thus its four $q \log c$ terms cancel in (ISM10); this proves the scalar invariance without altering any source form.

5.3 The canonical projector is exactly the original difference

Use $x \in [0, 1]$ for the comparison path, retaining the frequency y and all other deformation parameters. Write

$$M_x = (1 - x)M_0 + xM_1, \quad \dot{M} = M_1 - M_0, \quad X_x = M_x^{-1}\dot{M}. \quad (\text{ISM13})$$

Since $M_x X_x = \dot{M} = \dot{M}^*$, this is self-adjoint in the original M_x inner product. Define

$$\begin{aligned} P_N &= I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \\ Q_N &= B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0, \\ \Pi_N &= \rho_N G_N^{-1} \rho_N^* M_x. \end{aligned} \quad (\text{ISM14})$$

Each formula squares to itself and has M_x -self-adjoint adjoint law. The ranges of the first two are $\mathcal{P}_N$ and $B_N \mathcal{P}_{N-q}$; the range of the third is $\rho_N E$. By (ISM2)–(ISM3),

$$\mathcal{P}_N = \rho_N E \perp_{M_x} B_N \mathcal{P}_{N-q}. \quad (\text{ISM15})$$

To prove the equality, subtract $\rho_N \pi_\chi p$ from any $p \in \mathcal{P}_N$; (ISM2) puts the result in the relation range, and (ISM3) proves orthogonality. Therefore the original P_N, Q_N commute with product Q_N , and their orthogonal range decomposition proves the exact operator identity

$$\Pi_N = P_N - Q_N, \quad \text{Tr } \Pi_N = q. \quad (\text{ISM16})$$

The inverse of the range parametrization $\rho_N : E \rightarrow \text{ran } \Pi_N$ is the literal π_χ restricted to that range; both composites are identities by (ISM3). Thus (ISM16) identifies actual maps and their domains, not only their traces.

Define

$$\begin{aligned} \mathcal{A}_x &= \Pi_{q-1} + \Pi_q - \Pi_{2q-1} - \Pi_{2q}, \\ U_x &= Q_{2q-1} + Q_{2q} - Q_q, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{ISM17})$$

Substitution of (ISM16), with $Q_{q-1} = 0$, gives $\mathcal{A}_x = U_x - W_x$ and $\text{Tr } \mathcal{A}_x = 0$. These U_x, W_x are AW's U, W and SP's $\mathcal{R}, \mathcal{P}$, respectively. The source relation matrix is related to AW's unpadded matrix by $B_N = I_N B_N^{\text{AW}}$, so (ISM14) reproduces its actual projections with no sign or metric change.

Differentiating $\pi_x \rho_N = 1$ shows that ρ'_N has image in the original relation space. More explicitly, differentiation of $B_N^* M_x \rho_N = 0$ gives

$$\rho'_N = -B_N (B_N^* M_x B_N)^{-1} B_N^* \dot{M} \rho_N, \quad G'_N = \rho_N^* \dot{M} \rho_N. \quad (\text{ISM18})$$

The second identity follows because the two cross terms containing ρ'_N vanish by the same orthogonality. All these matrices are smooth on the closed interval, since their positive Gram inverses exist there. Jacobi differentiation and cyclicity of finite trace prove

$$\frac{d}{dx} \log V_N = \text{Tr}(X_x \Pi_N), \quad \mathcal{B}'(x) = \text{Tr}(X_x \mathcal{A}_x), \quad \mathcal{C} = \int_0^1 \text{Tr}(X_x \mathcal{A}_x) dx. \quad (\text{ISM19})$$

For the original row $v(y) = (1, S, \dots, S^{2q})$, $\Pi_N M_x^{-1} = \rho_N G_N^{-1} \rho_N^*$. Thus the signed density in PR29 is exactly

$$\kappa_x(y) = v(y) \mathcal{A}_x M_x^{-1} v(y)^*, \quad \int \kappa_x d\mu_x = \text{Tr } \mathcal{A}_x = 0. \quad (\text{ISM20})$$

The first equality keeps the inverse metric that is required by the original coordinate kernel. Integrating against the original signed measure $\mu_1 - \mu_0$ gives precisely (ISM19).

5.4 Restriction losses and the full cross trace

Fix $i \leq j$ among admitted degrees. The map $\rho_i - \rho_j : E \rightarrow \mathcal{P}_j$ has zero remainder, so (ISM2) makes it an original degree- j relation. Orthogonality of ρ_j therefore gives the cross-Grams

$$\rho_j^* M_x \rho_i = G_j, \quad \rho_i^* M_x \rho_j = G_j, \quad G_i - G_j = (\rho_i - \rho_j)^* M_x (\rho_i - \rho_j). \quad (\text{ISM21})$$

The exact reverse restriction on E is $T_{i,j} = G_i^{-1} G_j$. It is positive self-adjoint in G_i , with spectrum in $(0, 1]$, by the last identity. Cyclic trace and the first two identities give

$$\begin{aligned} \text{Tr}(\Pi_i \Pi_j) &= \text{Tr}(G_i^{-1} (\rho_i^* M_x \rho_j) G_j^{-1} (\rho_j^* M_x \rho_i)) \\ &= \text{Tr}(G_i^{-1} G_j), \\ \text{Tr}((\Pi_i - \Pi_j)^2) &= 2 \text{Tr}(1 - T_{i,j}). \end{aligned} \quad (\text{ISM22})$$

All traces are finite traces of the original coordinate operators. In particular their spectra are not declared to be arithmetic invariant summands.

Set

$$\begin{aligned} A_0 &= \Pi_{q-1} - \Pi_{2q-1}, \quad A_1 = \Pi_q - \Pi_{2q}, \\ \ell_0 &= \text{Tr}(1 - T_{q-1, 2q-1}), \quad \ell_1 = \text{Tr}(1 - T_{q, 2q}), \\ Z_x &= \text{Tr}(\mathcal{A}_x^2) = 2\ell_0 + 2\ell_1 + 2 \text{Tr}(A_0 A_1). \end{aligned} \quad (\text{ISM23})$$

This retains the cross term with its actual sign. On the full algebra $\text{End}(H)$, the map $L \mapsto M_x^{1/2} L M_x^{-1/2}$ is an algebra isomorphism with inverse $K \mapsto M_x^{-1/2} K M_x^{1/2}$. It sends the M_x -self-adjoint operators to Hermitian operators, and cyclicity preserves traces of every product. This proves $Z_x \geq 0$ and, from $\|A_0 + A_1\|_{\text{HS}}^2 \leq 2\|A_0\|_{\text{HS}}^2 + 2\|A_1\|_{\text{HS}}^2$, proves

$$0 \leq Z_x \leq 4(\ell_0 + \ell_1). \quad (\text{ISM24})$$

Here and below the Hilbert–Schmidt norm is computed after this exact metric isomorphism; no new source metric is used.

Put

$$c_x = \frac{\text{Tr } X_x}{n}, \quad \Sigma_x = \text{Tr}(X_x^2) - \frac{(\text{Tr } X_x)^2}{n}. \quad (\text{ISM25})$$

The trace is real by self-adjointness. Expanding $\text{Tr}((X_x - c_x I)^2)$ proves its equality with Σ_x , hence its nonnegativity. Trace zero of $\mathcal{A}_x$, and Cauchy–Schwarz for the two Hermitian images in the preceding isomorphism, give

$$|\mathcal{B}'(x)| \leq S_q(x) := \sqrt{\Sigma_x Z_x} \leq 2\sqrt{\Sigma_x(\ell_0 + \ell_1)}. \quad (\text{ISM26})$$

This bound can be combined pointwise with the existing full-spectrum, exact trace-norm, overlap, and angle bounds on the same $\mathcal{B}'$. It does not replace any smaller available bound.

5.5 A specified signed approximation and its exact residual

For an M_x -self-adjoint, continuous operator Y_x define $E_x = X_x - c_x I - Y_x$. It is self-adjoint too. Expanding the trace and applying the same Cauchy–Schwarz inequality proves the two signed inequalities

$$\text{Tr}(Y_x \mathcal{A}_x) - \sqrt{\text{Tr}(E_x^2) Z_x} \leq \mathcal{B}'(x) \leq \text{Tr}(Y_x \mathcal{A}_x) + \sqrt{\text{Tr}(E_x^2) Z_x}. \quad (\text{ISM27})$$

Every term is real. Integration is legitimate by continuity on the compact interval. We now construct such a Y_x , with its residual controlled from the actual two endpoint forms.

Let $D = M_0^{-1} M_1$, with ordered positive eigenvalues $b_1 \leq \dots \leq b_n$, counted with multiplicity. For this subsection write $B_x^\circ = (1-x)I + xD$, to distinguish it from every original relation matrix. Since $M_x = M_0 B_x^\circ$, direct inversion gives

$$X_x = (B_x^\circ)^{-1}(D - I), \quad \lambda_j(x) = 1 - x + x b_j, \quad \gamma_j(x) = \frac{b_j - 1}{1 - x + x b_j}. \quad (\text{ISM28})$$

Take an M_0 -orthogonal eigenbasis of D . It is also M_x -orthogonal, since on its j -th line M_x multiplies M_0 by $\lambda_j(x) > 0$. Therefore (ISM28) gives the full spectrum of X_x in the original varying metric.

Define the positive scalar and commuting operator

$$a_x = \frac{\lambda_1(x) + \lambda_n(x)}{2}, \quad H_x = I - B_x^\circ / a_x. \quad (\text{ISM29})$$

Their use does not rescale M_x , its moments, or its quotient Grams. The eigenvalues of H_x are $\zeta_j(x) = 1 - \lambda_j(x)/a_x$, and

$$\|H_x\|_{M_x} = \frac{x(b_n - b_1)}{2(1-x) + x(b_1 + b_n)} \leq \vartheta := \frac{b_n - b_1}{b_n + b_1} < 1. \quad (\text{ISM30})$$

Indeed the extrema of $1 - \lambda/a_x$ are equal and opposite at the two endpoints. The displayed scalar function of x has derivative $2(b_n - b_1)/(2(1-x) + x(b_1 + b_n))^2 \geq 0$. This proves the uniform bound, including $x = 0$.

For every integer $L \geq 0$, form the finite matrix polynomial and its centered version

$$\begin{aligned} X_x^{[L]} &= a_x^{-1} \sum_{r=0}^L H_x^r (D - I), \\ Y_x^{[L]} &= X_x^{[L]} - \frac{\text{Tr } X_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM31})$$

All factors commute with D and are real functions of its self-adjoint operator. They are consequently self-adjoint in both M_0 and M_x . The exact finite geometric-series identity $(I - H_x) \sum_{r=0}^L H_x^r = I - H_x^{L+1}$, together with $B_x^\circ = a_x(I - H_x)$, proves

$$\begin{aligned} R_x^{[L]} &:= X_x - X_x^{[L]} = H_x^{L+1} X_x, \\ E_x^{[L]} &:= X_x - c_x I - Y_x^{[L]} = R_x^{[L]} - \frac{\text{Tr } R_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM32})$$

No infinite-series interchange is used in this identity. In the original eigenbasis the residual eigenvalues are $\delta_j(x) = \zeta_j(x)^{L+1} \gamma_j(x)$. Thus its exact centered squared size is

$$\mathrm{Tr}((E_x^{[L]})^2) = \sum_{j=1}^n \delta_j(x)^2 - \frac{(\sum_{j=1}^n \delta_j(x))^2}{n}. \quad (\text{ISM33})$$

Both sides are nonnegative, and all multiplicities and signs of the residual eigenvalues are retained. The signed center is $\mathrm{Tr}(Y_x^{[L]} \mathcal{A}_x) = \mathrm{Tr}(X_x^{[L]} \mathcal{A}_x)$ by $\mathrm{Tr} \mathcal{A}_x = 0$. Equations (ISM27), (ISM31)–(ISM33) therefore provide a finite constructed signed center and its exact error.

There is also a uniform elementary error bound for this fixed original pair of forms. Define

$$K_D = \sum_{j=1}^n \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \quad (\text{ISM34})$$

The denominator of $\gamma_j(x)$ is at least $\min\{1, b_j\}$, so (ISM30)–(ISM33) imply $\mathrm{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$. The nested relation and source flags in (ISM17) give both U_x, W_x eigenvalues $0^{[q]}, 1^{[2]}, 2^{[q-1]}$: the first relation line contributes one, its next $q - 1$ increments contribute two, and the final increment contributes one; the source increments give the same list. In particular both have rank $q + 1$, and each dominates its range projection. Their ranges in dimension $2q + 1$ intersect in dimension at least one. The trace product of the two range projections is at least that intersection dimension, by summing squared projected lengths in an orthonormal basis. For positive operators, increasing a factor cannot decrease the product trace, since $\mathrm{Tr}(T^{1/2}(S' - S)T^{1/2}) \geq 0$. It follows that $\mathrm{Tr}(U_x W_x) \geq 1$. Using their complete spectra, we obtain

$$Z_x = 8q - 4 - 2 \mathrm{Tr}(U_x W_x) \leq 8q - 6. \quad (\text{ISM35})$$

This proof includes $q = 1$, where the multiplicity $q - 1$ is zero.

Set

$$\begin{aligned} J_L &= \int_0^1 \mathrm{Tr}(X_x^{[L]} \mathcal{A}_x) dx, \\ \epsilon_L &= \int_0^1 \sqrt{\mathrm{Tr}((E_x^{[L]})^2) Z_x} dx. \end{aligned} \quad (\text{ISM36})$$

Every integrand is a specified continuous finite matrix expression on the original source path. Integrating (ISM27) with the constructed (ISM31) and the established uniform bound gives

$$J_L - \epsilon_L \leq \mathcal{C} \leq J_L + \epsilon_L, \quad 0 \leq \epsilon_L \leq \vartheta^{L+1} \sqrt{K_D(8q - 6)} \xrightarrow{L \rightarrow \infty} 0. \quad (\text{ISM37})$$

Thus the supplied signed enclosure now has an explicit convergent residual on every fixed original finite source. It retains the arithmetic correlation in its center. The evaluation of the center integral still uses the specified arithmetic entries and a controlled integration method; this calculation supplies no uniform estimate in k . If $b_n = b_1$, then $D = b_1 I$, $H_x = 0$, the residual and $\mathcal{C}$ vanish exactly, and every formula above remains valid.

5.6 The original primitive and the residual at a proper support

We finally prove the source comparison underlying all these metric changes. In the original one-factor complex $V \xrightarrow{\Theta} \mathcal{B}$, retain $h(D)F_h = \Theta \phi_*$, with the original D action commuting with Θ . For $v \in E$, the polynomial $(\rho_1 - \rho_0)v = \chi Q_v$ has the coefficient $Q_v = (Z_N(M_0) - Z_N(M_1))v$ by (ISM4). In the multivariate polynomial ring put

$$f_v(s_1, \dots, s_k) = \chi(s_1 + \dots + s_k) Q_v(s_1 + \dots + s_k). \quad (\text{ISM38})$$

Its image in $E_h^{\otimes k}$ is zero, because $\chi(A_h^{(k)})1 = 0$. Successive monic division in $s_1, \dots, s_k$ constructs specific polynomials $Q_{v,i}$ as follows. Let $f_{v,0} = f_v$ and write $f_{v,i-1} = h(s_i)Q_{v,i} + f_{v,i}$, where $\deg_{s_i} f_{v,i} < \deg h$. Earlier remainder degree bounds are preserved because $h(s_i)$ uses only s_i . The final remainder $f_{v,k}$ has degree below $\deg h$ in every variable and zero class in the quotient, so it is zero by the tensor monomial basis. Telescoping proves

$$f_v = \sum_{i=1}^k h(s_i)Q_{v,i}. \quad (\text{ISM39})$$

This constructs a primitive using precisely the original relations, without replacing them by a larger observation kernel.

In the tensor complex $\mathbf{C}^\bullet = [V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ define

$$K_v = \sum_{i=1}^k (-1)^{i-1} Q_{v,i}(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}) \in \mathbf{C}^{k-1}. \quad (\text{ISM40})$$

The displayed source tensor has exactly $i-1$ preceding factors of degree one. Its differential therefore has sign $(-1)^{i-1}$; multiplying by the sign in (ISM40) gives $+1$. The source operators commute with the differential. Applying $h(D_i)F_h = \Theta\phi_*$ to (ISM39) now proves

$$dK_v = \mathcal{V}((\rho_1 - \rho_0)v). \quad (\text{ISM41})$$

Consequently the two canonical sections have the same class in the original full-source quotient; their common arithmetic observation is exactly $\eta_{h,k}v$ by (ISM1). Composing with the original full-source quotient $q^{(k)}$ gives the already constructed map $\sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}$. Here no map was made natural merely by changing its metric; the actual primitive (ISM40) is the comparison.

To express the full two-leg source without deleting its diagonal, retain the labelled Fourier copy $\widehat{V}$, with the original isomorphism $\mathcal{F} : V \rightarrow \widehat{V}$, and the complex $\mathbf{T}^\bullet = [V \oplus \widehat{V} \rightarrow \mathcal{B}]$, whose differential is $(\phi, \widehat{\psi}) \mapsto \Theta(\phi - \mathcal{F}^{-1}\widehat{\psi})$. There are exact cochain maps from the one-leg complex to this complex with degree-one map identity and degree-zero maps

$$j_+(v) = (v, 0), \quad j_-(v) = (0, -\mathcal{F}v).$$

Indeed either differential equals Θv . The reverse cochain map has degree-one map identity and degree-zero map $p(\phi, \widehat{\psi}) = \phi - \mathcal{F}^{-1}\widehat{\psi}$. Thus $pj_+ = pj_- = 1$, whereas $(j_+ - j_-)(v) = (v, \mathcal{F}v)$ is the full degree-zero diagonal kernel. These formulas prove both composites and retain that kernel explicitly. Tensoring specified choices of these degree-zero maps with their degree-one identities gives cochain maps, since the displayed differential identities preserve each original tensor sign. Apply one such map to (ISM40) to obtain $\widetilde{K}_v \in (\mathbf{T}^{\otimes k})^{k-1}$. Its top differential is still (ISM41).

For precision at a proper declared support $\lambda = (W, S; A)$, let $\mathcal{J}$ be the specified finite carrier of coefficient positions with $A \subseteq \mathcal{J}$. Put $\mathbf{C}_{\text{full}}^\bullet = (\mathbf{T}^{\otimes k})^{\oplus \mathcal{J}}$, with the original differential on each coordinate. Insert the active A -coordinates by literal zero insertion into this fixed full carrier. Let $\mathbf{C}_\lambda^\bullet \subseteq \mathbf{C}_{\text{full}}^\bullet$ be its declared subcomplex, retaining its independent outer label and coefficient mask. The same source primitive is taken in the indicated coefficient copy. When the compared top source representatives lie in $\mathbf{C}_\lambda^k$, their difference has the exact residual

$$[d\widetilde{K}_v]_\lambda \in \frac{d\mathbf{C}_{\text{full}}^{k-1} \cap \mathbf{C}_\lambda^k}{d\mathbf{C}_\lambda^{k-1}}. \quad (\text{ISM42})$$

This quotient is the kernel of the natural map $\mathbf{C}_\lambda^k/d\mathbf{C}_\lambda^{k-1} \rightarrow \mathbf{C}_{\text{full}}^k/d\mathbf{C}_{\text{full}}^{k-1}$: a class maps to zero exactly when its representative is a full-source boundary. The inclusion of the displayed numerator therefore induces the kernel isomorphism, with inverse obtained by using the same representative. These two maps are well-defined since differences in $d\mathbf{C}_\lambda^{k-1}$ are precisely the equivalence relation on both sides.

Thus (ISM41) kills the class at this support exactly when $d\tilde{K}_v \in d\mathcal{C}_\lambda^{k-1}$. Membership $\tilde{K}_v \in \mathcal{C}_\lambda^{k-1}$ is one explicit sufficient witness, not an assumption made for arbitrary W .

First specialize to $k = 1$ and the single coefficient carrier $\mathcal{I} = \{i_0\}$, with active face $A = \{i_0\}$. The coordinate projection onto i_0 and its inverse coordinate insertion identify this one-coordinate carrier with the original one-factor complex. In these specified coordinates put $\mathcal{B}_\lambda = \mathcal{C}_\lambda^1 \subseteq \mathcal{B}$. Suppose the admitted source image under p is $W \subseteq V$, so its boundary range is exactly ΘW , and retain $\Theta V \subseteq \mathcal{B}_\lambda$. The precise kernel map in this single coordinate is

$$V/W \longrightarrow \mathcal{B}_\lambda/\Theta W, \quad [v] \longmapsto [\Theta v]_W. \quad (\text{ISM43})$$

It is injective because Θ is injective: $\Theta v = \Theta w$, $w \in W$, forces $v = w$. Its range is the kernel of the map to the full quotient $\mathcal{B}/\Theta V$, since being in that kernel means being represented by Θv . In this case (ISM42) is exactly the class of the original one-factor primitive in V/W .

For an arbitrary admitted face $A \subseteq \mathcal{I}$, restore the original coordinate injections $\iota_a : \mathcal{B} \rightarrow \mathcal{B}^{\oplus \mathcal{I}}$ and projections $\text{pr}_a : \mathcal{B}^{\oplus \mathcal{I}} \rightarrow \mathcal{B}$. They satisfy $\text{pr}_b \iota_a = \delta_{ab} 1$ and $\sum_{a \in A} \iota_a \text{pr}_a = 1$ on the A -supported subspace. Let the degree-one source at this face be $\mathcal{B}_\lambda^{\oplus A}$, inserted in the fixed carrier by these ι_a , and retain the degree-zero source image $W^{\oplus A}$ and boundary $(\Theta W)^{\oplus A}$. Its kernel map is

$$(V/W)^{\oplus A} \longrightarrow \mathcal{B}_\lambda^{\oplus A}/(\Theta W)^{\oplus A}, \quad ([v_a])_{a \in A} \longmapsto \left[\sum_{a \in A} \iota_a \Theta v_a \right].$$

Replacing v_a by $v_a + w_a$, $w_a \in W$, changes the displayed sum by the specified boundary, so the map is well-defined. If its image is zero, projection by each pr_a gives $\Theta v_a \in \Theta W$, whence $v_a \in W$ by injectivity of Θ . Thus the map is injective. A representative in its codomain maps to zero in $(\mathcal{B}/\Theta V)^{\oplus \mathcal{I}}$ exactly when each active coordinate is Θv_a for some $v_a \in V$. Its inverse on that kernel takes these unique preimages modulo W . Changing a representative by an element of $(\Theta W)^{\oplus A}$ changes each preimage by an element of W . The coordinate identities above then prove both inverse composites. This is the exact $(V/W)^{\oplus A}$ residual, with its full ambient carrier specified.

For $A \subseteq A'$, zero insertion sends the tuple $([v_a])_{a \in A}$ to the same tuple extended by zero; both orders of this insertion and the kernel map give $\sum_{a \in A} \iota_a \Theta v_a$. A signed coefficient permutation sends that sum to $\sum_{a \in A} \epsilon_a \iota_{\sigma(a)} \Theta v_a = \sum_{a \in A} \iota_{\sigma(a)} \Theta(\epsilon_a v_a)$, which proves the same commuting identity with every sign retained. When $A = \emptyset$, the two displayed direct sums and their unique map are zero vector spaces and a zero map at the retained outer label $(W, S; \emptyset)$; the coordinate calculation does not identify that label with external absence. This proves the typed relation to the existing mixed-support quotients while retaining every proper-source residual.

All source identities, projector equalities, scaled comparison signs, and signed estimates above use the same original finite polynomial presentation. Their full primitive and proper support kernel have now been exhibited, so backward and forward propagation can replace weaker metric bounds at their earlier use sites without erasing the older sources or claiming that the arbitrary proper-support residual vanishes.

6 Earlier conclusion replacement blocks

6.1 R44 endpoint costs

Both additional logarithms are nonnegative by minimization over nested lift spaces. Substitution of $L_{h,k}/q_k \geq \delta k/2$, division by $q_k \log k$, and $(q_k - 1)/q_k \rightarrow 1$ prove the necessary four-volume lower limit 4 in (EP.50e). The new signed control is now substituted into this same identity. Write $q = q_k$ and retain the arithmetic endpoint losses $E_0 = \log(V_{q-1}/V_q)$, $E_1 = \log(V_{2q-1}/V_{2q})$, and put $\mathcal{B}_{h,k} = 2C_k + E_0 + E_1$. On the original Gamma-to-arithmetic moment path, (FV.M1)–(FV.M18) proves the

exact identity $\mathcal{B}_{h,k} = \mathcal{B}_{h,k}^\Gamma + \Delta_{h,k}$ and $|\Delta_{h,k}| \leq \mathfrak{J}_{h,k}$, where

$$\mathfrak{J}_{h,k} = \int_0^1 \min\{S_{\text{AW}}(x), T_q(x), S_{\text{SP}}(x), S_{\text{HS}}(x), (2q-1)\sqrt{1 - e^{-F(x)/(2q-1)}} d_x\} dx.$$

Here $F(x)$ is the unchanged four-volume function and d_x the full original relative spectral diameter. The full formulas for S_{AW} , S_{SP} , their exact source correspondence and the nonlinear coordinate are proved at that original endpoint site. Equation (FV.M15a) retains $T_q = d_x \|U_x - W_x\|_1/2$ and $S_{\text{HS}} = \sqrt{\Sigma_x \text{Tr}((U_x - W_x)^2)}$, where $\Sigma_x = \text{Tr}(C_x^2) - (\text{Tr } C_x)^2/(2q+1)$, and proves both bounds on the original metric by midpoint centering and centered Hilbert–Schmidt Cauchy–Schwarz. The last minimum entry is the actual-moment specialization of RMT11. Every entry is retained, including the exact T_q and its majorant S_{SP} . Let $a = 2q - 1$, $H_0 = 2 \text{arcosh}(e^{\mathcal{B}_{h,k}^\Gamma/(2a)})$, and retain the complete $\mathfrak{J}_{h,k}^{\text{nl}}$ of (FV.M15). For every finite truncation $p_N \geq 0$, the exact source dictionary and finite construction in (FV.M16a)–(FV.M16c) give the signed center and residual j_{p_N}, e_{p_N} : these are exactly J_L, ϵ_L of ISM36 with $L = p_N$, and $X_x^{\text{ISM}} = C_x$, $\mathcal{A}_x^{\text{ISM}} = U_x - W_x$. They obey $j_{p_N} - e_{p_N} \leq \Delta_{h,k} \leq j_{p_N} + e_{p_N}$ and $e_{p_N} \leq \vartheta^{p_N+1} \sqrt{K_D(8q-6)} \rightarrow 0$ for the fixed original pair of forms, with the unchanged ϑ, K_D in (FV.M16b). Their integrals and the source’s own integration errors remain explicit. The simultaneous endpoints are

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{lo}} &= \max\left\{0, \mathcal{B}_{h,k}^\Gamma - \mathfrak{J}_{h,k}, 2a \log \cosh \left(\frac{\max\{0, H_0 - \mathfrak{J}_{h,k}^{\text{nl}}\}}{2} \right), \right. \\ &\quad \left. \mathcal{B}_{h,k}^\Gamma + j_{p_N} - e_{p_N} \right\}, \\ \mathcal{B}_{h,k}^{\text{hi}} &= \min\left\{ \mathcal{B}_{h,k}^\Gamma + \mathfrak{J}_{h,k}, 2a \log \cosh \left(\frac{H_0 + \mathfrak{J}_{h,k}^{\text{nl}}}{2} \right), \right. \\ &\quad \left. \mathcal{B}_{h,k}^\Gamma + j_{p_N} + e_{p_N} \right\}. \end{aligned}$$

Monotonicity of the exact inverse coordinate in (FV.M16)–(FV.M18) and the signed bound give

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{lo}} &\leq \mathcal{B}_{h,k} \leq \mathcal{B}_{h,k}^{\text{hi}}, \\ 4(q-1) \log \frac{L_{h,k}}{C_h^{\text{bal}} q} + E_0 + E_1 &\leq \mathcal{B}_{h,k}^{\text{hi}}, \quad C_k \leq \frac{\mathcal{B}_{h,k}^{\text{hi}} - E_0 - E_1}{2}. \end{aligned}$$

The last two inequalities substitute the retained central lower bound and the exact four-volume identity; neither arithmetic endpoint loss is omitted. The proper-source primitive residual is the exact tensor intersection quotient (FV.M13), with the one-leg V/W map proved in (FV.M14), so this metric comparison does not erase any proper-source class.

6.2 R45 upper gap

All constants and the literal source mass are retained. For $\mathcal{L}_k = 2(q_k - 1) \log(\delta k / (2C_h^{\text{bal}}))$, (AU.32) computes every term of $\mathcal{U}_k - \mathcal{L}_k$, and (AU.33) proves that its quotient by q_k^2 tends to $\alpha D + \log 256 > 0$. This upper estimate leaves the stated gap to the quartet lower estimate. The propagated original-source estimate is the additional bound

$$C_k \leq \mathcal{U}_k^{\text{signed}} := \frac{\mathcal{B}_{h,k}^{\text{hi}} - E_0 - E_1}{2}.$$

Its complete proof (AU.31a)–(AU.31b) transports the unchanged monomial source and original relation by $P(S) \mapsto P(k/2 + iu)$, retaining the factor i^q and every raw derivative factor i^d . This signed entry uses actual moments through $4q$; the original first matrix bound continues to require actual moments only through $2q$. Let $\hat{\mathcal{U}}_k$ be the minimum of this signed entry and all the full bounds in

(AU.31b), including the original $\mathcal{U}_k$ and the actual low-degree matrix entry. They bound the same C_k , so

$$\mathcal{L}_k \leq C_k \leq \hat{\mathcal{U}}_k \leq \mathcal{U}_k, \quad 0 \leq \hat{\mathcal{U}}_k - \mathcal{L}_k \leq \mathcal{U}_k - \mathcal{L}_k.$$

The positive limit in (AU.33) belongs to the explicitly displayed old scalar majorant; it is not assigned to this refined minimum. The complete angle, spectral, and overlap integrands remain the actual quantities controlling the newly narrowed finite interval.

6.3 R47 joint budget

The arithmetic remainder estimate is extended to every admitted degree with its entire monic quotient formula. In the original notation of the upper-comparison chapter, it gives

$$G_{q-1} \preceq C_N G_N, \quad C_N = \frac{U_k(T) A_N^2 B_q(r)^2}{T a_k(T)}, \quad N \geq q-1,$$

with $T = Dq$, $r = kR_0/T < 1$, the unchanged original density mass, and all constants defined in the complete proof. Thus the actual endpoint gaps C_{2q-1}^{-1} and C_{2q}^{-1} are proved arithmetic inequalities. The exact ratio satisfies $C_{2q}/C_{2q-1} \leq 147/5 < 32$, so the same construction gives $\mathcal{B}_{h,k} \leq 2\mathcal{U}_k + q \log 32$ with $\mathcal{U}_k$ from R45. The same-source metric refinement now gives the stronger literal joint budget

$$\mathcal{B}_{h,k}^{\text{lo}} \leq \mathcal{B}_{h,k} \leq U_{h,k}^{\text{joint}} := \min\{q \log C_{2q-1} + q \log C_{2q}, \mathcal{B}_{h,k}^{\text{hi}}\}.$$

Equation (ERJ.32a) proves the required equality of the monic-source and monomial-source quotient kernels before taking this minimum. The complete five-estimate pointwise minimum and its nonlinear transport therefore apply to these exact original restriction returns. The two finite trace-certificate upper endpoints below may also be summed and included in the same minimum; their original entry errors and series tails remain attached to their actual matrices.

6.4 R47 inverse powers

Indeed the central trace is one plus the inverse-power sum of its remaining $q-1$ eigenvalues. The tangent inequality $e^y \geq 1+y$ at their logarithmic mean bounds this sum from below by their geometric mean, and the ordered comparison transfers every eigenvalue. These same inverse observations also acquire the finite upper control proved in (ERJ.32b):

$$\text{Tr } U_{q-1,2q-1}^{-s} + \text{Tr } U_{q,2q}^{-s} \leq 2q-1 + e^{s\mathcal{B}_{h,k}} \leq 2q-1 + e^{sU_{h,k}^{\text{joint}}}.$$

Indeed all $2q$ return eigenvalues $g_{a,j}$ lie in $(0, 1]$ and their negative logarithms sum exactly to $\mathcal{B}_{h,k}$. Writing $z_{a,j} = g_{a,j}^{-s} - 1 \geq 0$, expansion gives $\prod(1+z_{a,j}) \geq 1 + \sum z_{a,j}$, which is precisely the displayed bound. Each individual eigenvalue also satisfies $g_{a,j} \geq e^{-\mathcal{B}_{h,k}} \geq e^{-U_{h,k}^{\text{joint}}}$. The original lower bounds and central unit eigenvalues above are retained simultaneously, with their full phase/contraction penalty.